Modeling and Forecasting Emerging Events across Space and Time

Ali Arab

Georgetown University
Professor

Department of Mathematics & Statistics
University of Maryland, Blatimore County
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Outline

- Challenges of Data sparsity and modeling dynamics
- Motivating examples:
  - (Re)emerging epidemics, Spread of rare conditions (e.g., Lyme disease)
  - Conflict- or climate-driven forced migration
- Modeling strategy
  - Zero-modified models
  - Mechanistic models
- Understanding drivers of dynamics
- Forecasting and Transfer Learning

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(Re)Emerging Epidemics/Crises

Data sparsity is one of the main challenges of modeling emerging events (epidemics, crises, etc.) and spread of rare conditions (low number of cases over space and time). Of course, detectability is also an issue in these cases. Generally, the modeling issues are similar to modeling **Rare Events** in many regards:

- low frequency, low probability, or “unexpected”
- relative rareness over time and/or space
- difficult to understand and model spatio-temporal dynamics

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Modeling Challenges & Approaches

Some of the challenges of these type of data include:

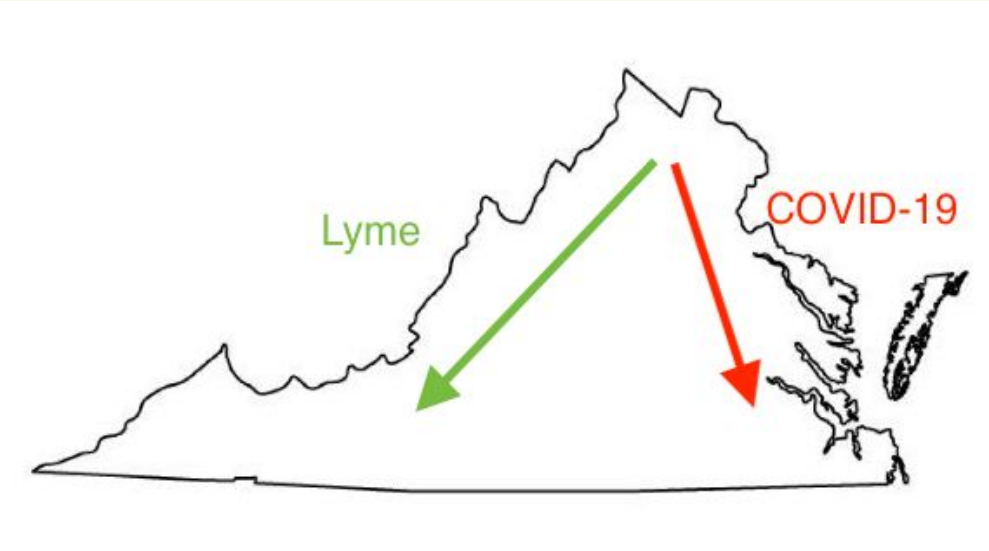
- Excess zeroes
- Small sample size of non-zero values (and typically spatially clustered)
- Large range of values: many zeroes, few non-zeroes, sometimes very few very large values (heavy tailed)
- complex dynamics over space and/or time (e.g., infectious diseases spread and become more common in areas that it used to be rare).

In addition to excess zeros and possibly heavy tails, modeling challenges include understanding the drivers and mechanism of the dynamics, and address spatial and temporal variability in the data.

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Motivating Problems: Spread of Lyme & COVID-19 in VA

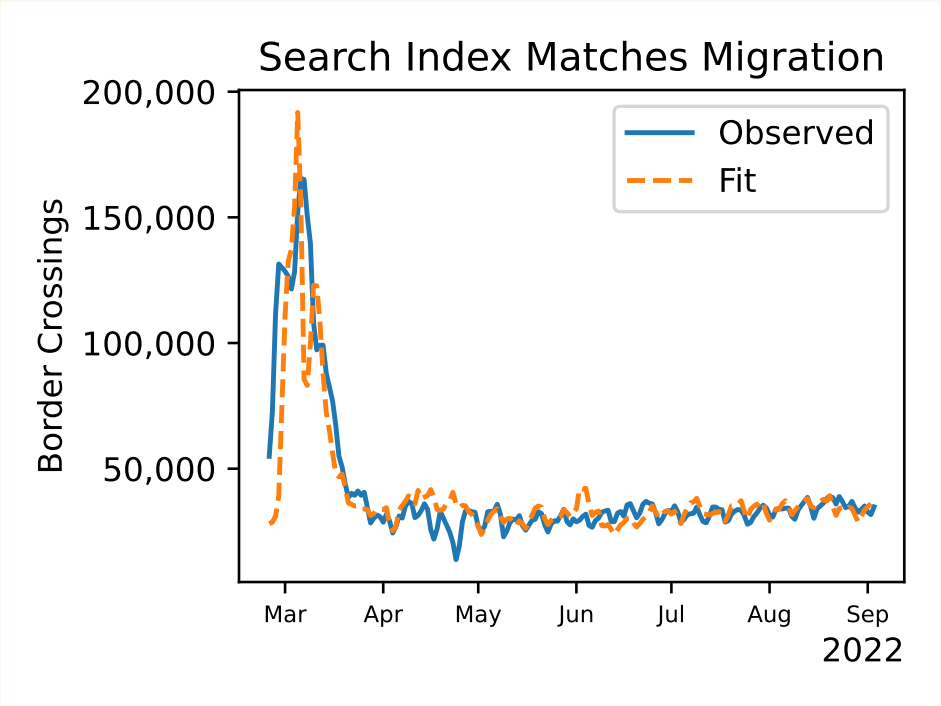
Difference in the geographical direction of the spread of Lyme and COVID-19 in Virginia!



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Motivating Problems: Forced displacement in Ukraine

Border crossings to Poland, Slovakia, and Hungary. Google trends indicator: search volume of Ukrainian terms related to travel (e.g., plane tickets, land routes, etc.).



Wycoff et al. (2024): The digital trail of Ukraine's 2022 refugee exodus.

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Modeling Framework

Goal: Develop a modeling framework to address the need for modeling and forecasting emerging events and crises.

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Bayesian Hierarchical Model

Basic Hierarchical Model (Berliner 1996)

1. [data|process, parameters]
2. [process|parameters]
3. [parameters]

Using the Bayes Theorem, the posterior [process,parameters|data] can be written as proportional to the product of these three distributions!

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Data Model Choice

Common choices for modeling these data include zero-modified models (See Arab 2015 for a discussion) including zero-inflated and hurdle models (i.e., mixture of a zero generating process and a count model), For example, a Poisson hurdle model

$$(1 - p_{i,t}) I_{(y_i = 0)} + p_{i,t} \text{Poi}(\lambda_{i,t} > 0),$$

or a negative binomial hurdle model:

$$(1 - p_{it}) \mathbf{1}(y_{i,t} = 0) + p_{i,t} f_{\text{ZTNB}}(y_{i,t} | r, q)$$

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Data Model Choice

For a discussion of zero-modified models, see:



Arab (IJERPH, 2015): Spatial and Spatio-Temporal Models for Modeling Epidemiological Data with Excess Zeros.

Other (less common) choices include models for heavy-tailed data, and models that address both heavy tails and excess zeroes such as a double hurdle model (e.g., Balderama et al. 2016). These models may be preferred when prediction of extreme cases is of interest.

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Modeling the Dynamics

Dynamics of emerging epidemics over space and time are complex and are often modeled using **autoregressive** models which may have limited capacity to mimic the true dynamics.

Alternative approaches include physical-statistical models such as **PDE-based models** (Wikle 2003), or **agent based Models** (Hooten and Wikle 2010).

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Motivating Problem

Modeling the spread of vector-borne diseases such as Lyme disease

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Lyme Disease

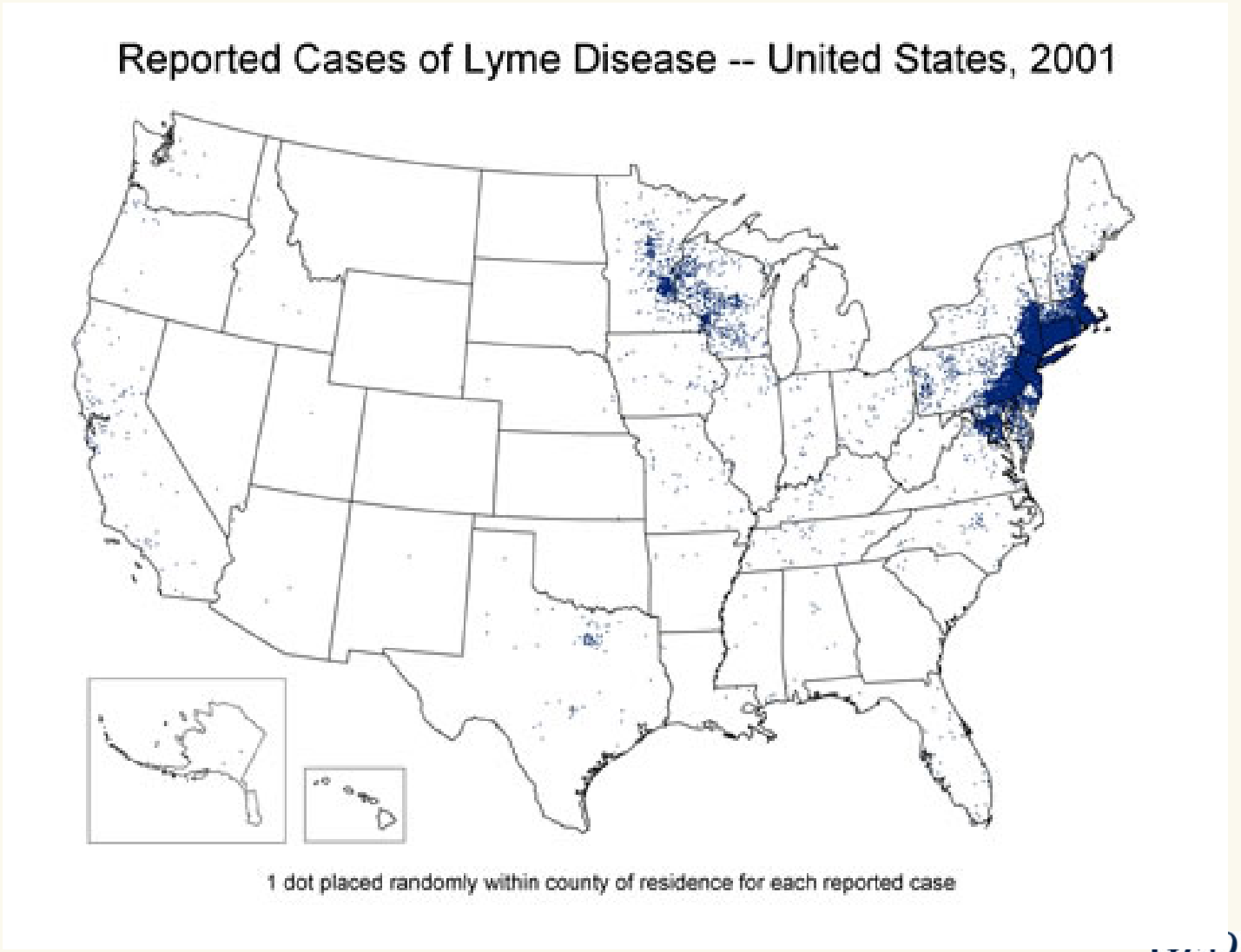
First identified in Lyme, Connecticut, Lyme disease is a bacterial illness that can cause fever, fatigue, joint pain, and skin rash, as well as more serious joint and nervous system complications. It is the most common vector-borne disease in the United States.

Lyme is common in parts of the upper East Coast with a high incidence rate but uncommon in other areas (e.g., almost non-existent in Arkansas).

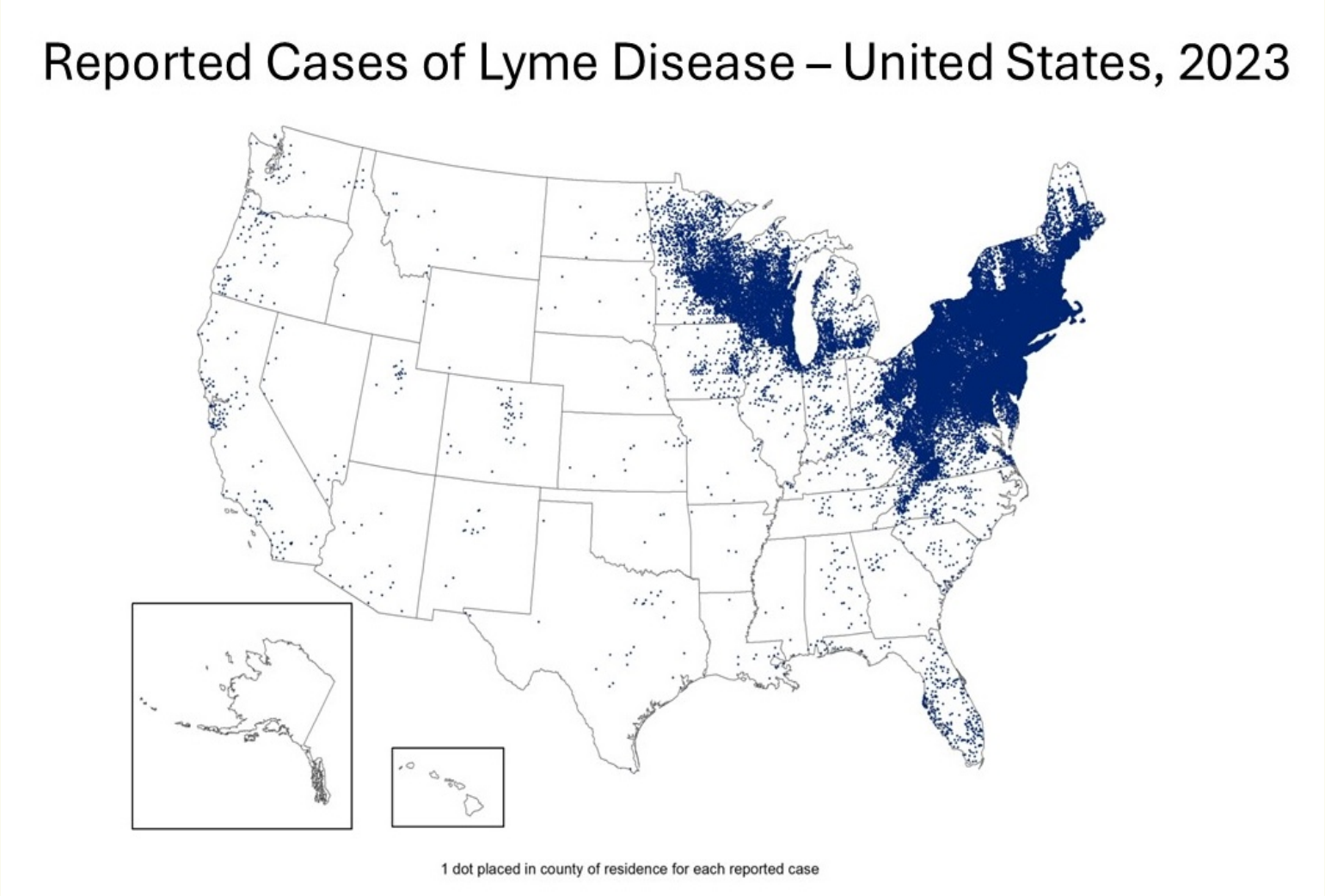
“The incidence of Lyme disease in the United States has approximately doubled since 1991, from 3.74 reported cases per 100,000 people to 7.95 reported cases per 100,000 people in 2014.” (EPA site on Climate Change Indicators)

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Reported Lyme cases, 2001–2016



Reported Lyme Cases, 2023

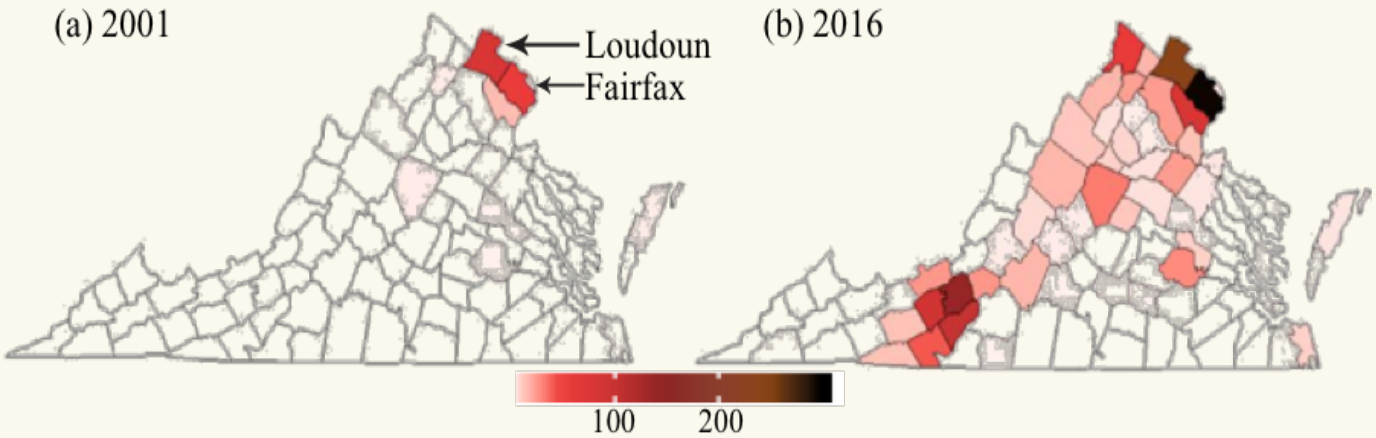


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Lyme Disease in Virginia

In a previous study, we analyzed cases of Lyme in Virginia during 2001-2016.

Annual number of confirmed cases of Lyme at the county level (Source: CDC)



Here, we will consider 2001-2023 data.

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Data Model

Let the random variable $Y_{i,t}$ represent the count of the disease:

$$[Y_{i,t}|\mu_{i,t}, p_{i,t}] \sim \text{NegBinHurdle}(\mu_{i,t}, p_{i,t}).$$

where the negative binomial hurdle model is defined as

$$p_{i,t} I_{(y_i=0)} + (1 - p_{i,t}) \text{ NegBin}(\mu_{i,t} > 0).$$

Notation:

$$\mathbf{Y}_t = (Y_{1,t}, \ldots, Y_{n,t})',$$

$$\mathbf{p}_t = (p_{1,t}, \ldots, p_{n,t})'.$$

$$\boldsymbol{\mu}_t = (\mu_{1,t}, \ldots, \mu_{n,t})'.$$

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Process Model

We consider generalized additive models for two process models: a log-linear model for the mean of a zero-truncated negative binomial:

$$\log(\boldsymbol{\mu}_t) = \mathbf{X}\boldsymbol{\alpha} + \mathbf{w}_s + \boldsymbol{\eta}_t.$$

and a logistic regression model for the presence probabilities

$$\text{logit}(\mathbf{p}_t) = \mathbf{X}_p\boldsymbol{\beta} + \boldsymbol{\eta}_s + \boldsymbol{\varepsilon}_t,$$

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Predictor Data

We consider monthly values of the following environmental factors:

- Surface temperature,
- Relative humidity,
- Precipitation,
- Wind components (V-component along latitude and U-component along longitude) from the North American Regional Reanalysis;
- Vegetation index (NDVI).

We also consider elevation and population (offset).

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Parameter Models

In the Bayesian setting, the parameter models for a hierarchical model are the priors!

We define relatively non-informative prior densities for the unknown parameters (e.g., normal distribution with mean 0 and large variance; inverse-Gamma distributions with small mean and large variance for variance components).

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Drivers of Dynamics

However, this model does not have any components to effectively capture the dynamics.

Let's focus on understanding the drivers of dynamics.

Understanding Drivers of Dynamics

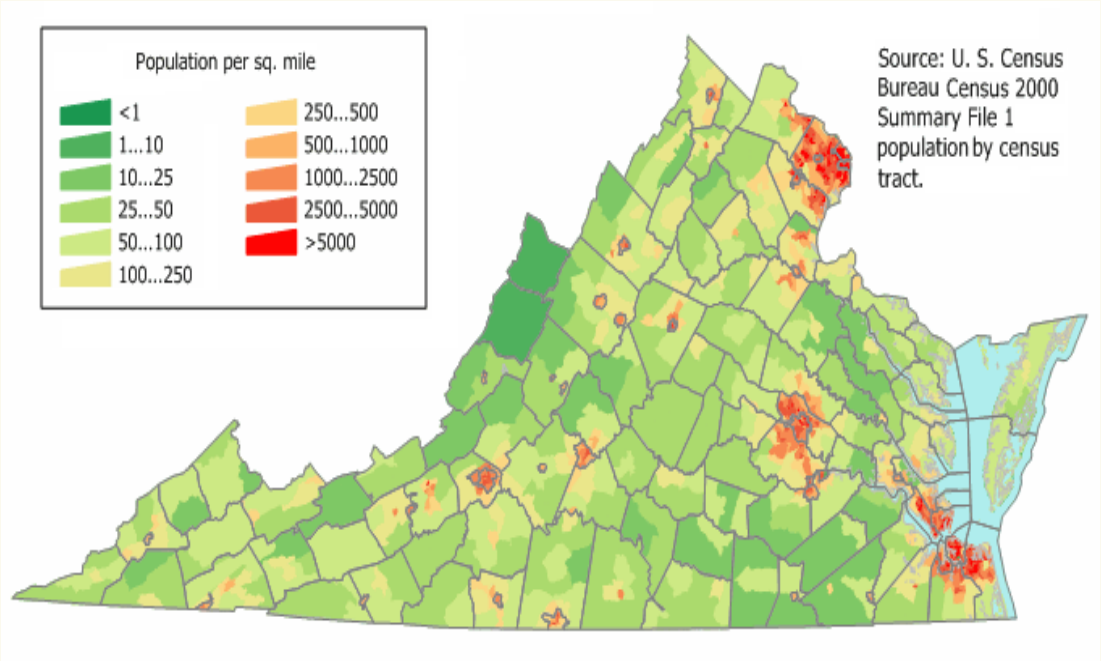
An important factor to consider in modeling the diseases dynamics is the mechanism of disease spread and its drivers. The drivers of dynamics of disease spread are often specific to classes of infectious diseases and may even be disease-specific.

For example, the dynamics of disease spread for **vector-borne diseases** are often a function of the environment (and ecology), and human behavior (and mobility). While the dynamics in **human-to-human transmission** is largely a function of human behavior, population and mobility.

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Drivers of Dynamics

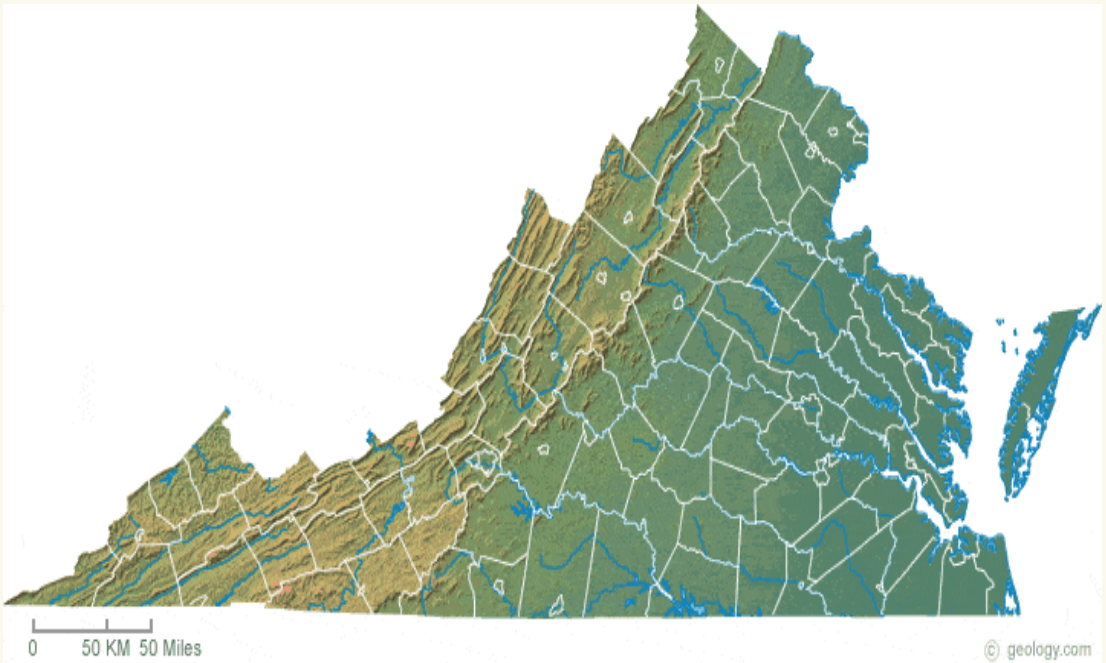
Population Density



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Drivers of Dynamics

Environmental factors such as elevation, greenness, wind, temperature, and precipitation.



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Modeling

So there are opportunities for spatio-temporal learning.

...But how can we go about modeling and forecasting the dynamics at the early stages of spread of the disease?

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Mechanistic PDE-Based Process Model

Mechanistic (physical-statistical) models may offer a reasonable solution.

For example, the presence/absence probabilities can be modeled as:

$$\text{logit}(\mathbf{p}_t) = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}_t + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{u}_t = (u_{1,t}, \ldots, u_{n,t})'$ is a latent process characterized based on a diffusion PDE, and ε_t's are *iid* error terms (due to conditional independence):

$$\boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}).$$

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Process Model: PDE

The latent process will be characterized based on the following two-dimensional diffusion equation:

$$\frac{\partial u_t(x,y)}{\partial t} - \theta\left(\frac{\partial^2 u_t(x,y)}{\partial x^2} + \frac{\partial^2 u_t(x,y)}{\partial y^2}\right) = 0.$$

where θ denotes the diffusion parameter.

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Process Model

Using a forward differencing time discretization:

$$\frac{\partial u_t}{\partial t} = \frac{u_t - u_{t-\Delta t}}{\Delta t},$$

and centered differences in space:

$$\frac{\partial^2 u_t}{\partial x^2} = \frac{u_t(x+\Delta x, y) - 2u_t(x,y) + u_t(x-\Delta x, y)}{\Delta^2 x},$$

$$\frac{\partial^2 u_t}{\partial y^2} = \frac{u_t(x, y+\Delta y) - 2u_t(x,y) + u_t(x, y-\Delta y)}{\Delta^2 y},$$

the system of equations may be discretized and presented as a dynamical model, $\mathbf{u}_t = \mathbf{H}\mathbf{u}_{t-1}$, where the transition matrix H is **sparse** and only a function of θ, Δt, Δx, and Δy.

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Process Model

Finally, the process model includes the discretized PDE with additive error:

$$\text{logit}(\mathbf{p}_t) = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}_t + \boldsymbol{\varepsilon}_t,$$

$$\mathbf{u}_t = \mathbf{H}\mathbf{u}_{t-1} + \boldsymbol{\eta}_t,$$

where $\boldsymbol{\eta}_t \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ and the variance-covariance matrix $\boldsymbol{\Sigma}$ is parameterized to reflect spatial dependence among the locations:

$$\boldsymbol{\Sigma} = \sigma^2 \mathbf{R}(\tau),$$

$$\mathbf{R}(\tau) = \exp(-\tau ||d||)$$

$\mathbf{R}(\tau)$ represents the spatial correlation based on the Euclidean distance between locations ($||d||$). Many other choices!

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Model Implementation & Results

Model Implementation is based on Markov Chain Monte Carlo (MCMC):

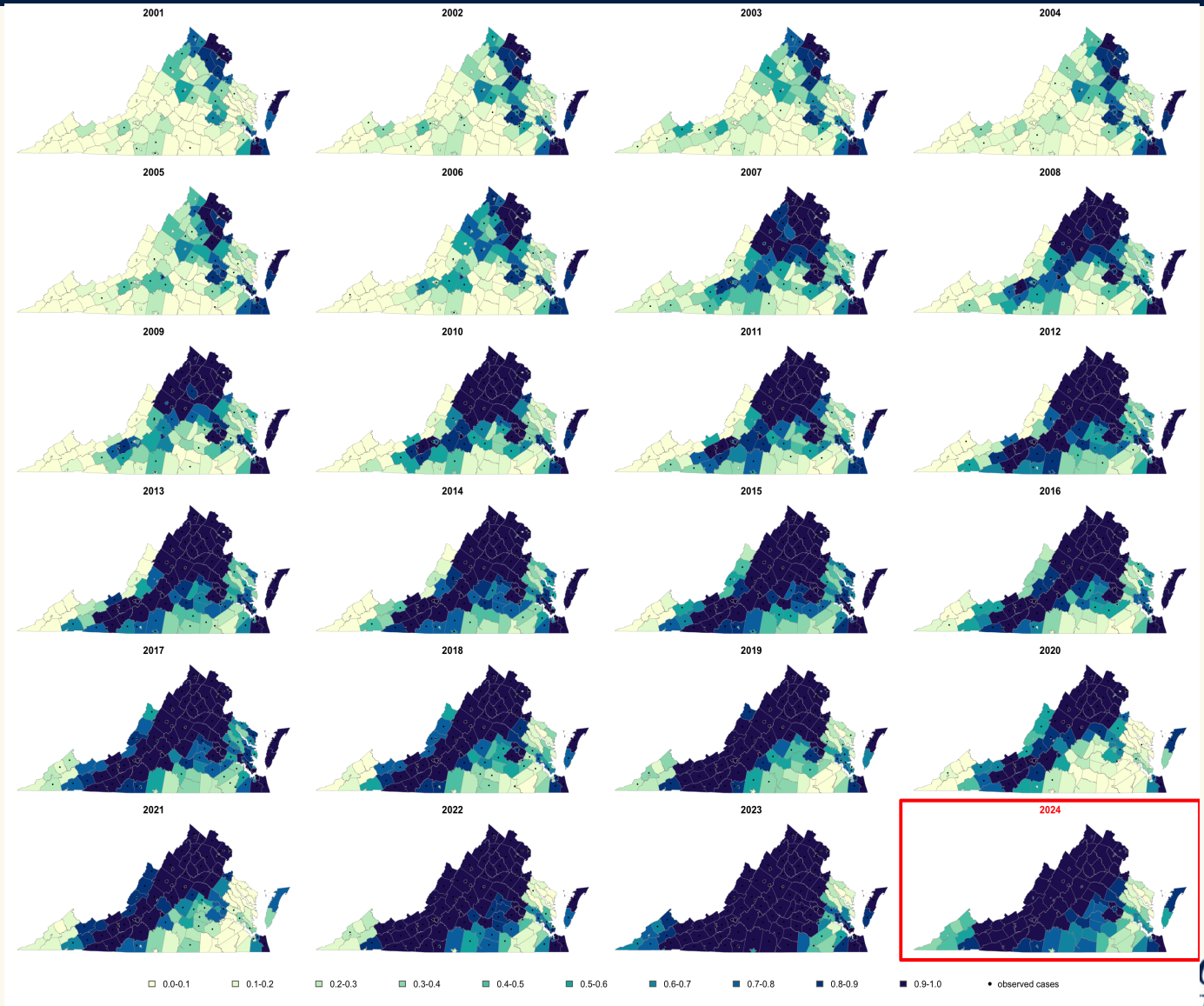
- Metropolis-Hastings within Gibbs Sampling,
- 4 chains, each 20,000 iterations, and 5000 burn-in.

Parameter	**Post.Mean**	**95%CI**
θ	0.0025	(0.0009, 0.0051)

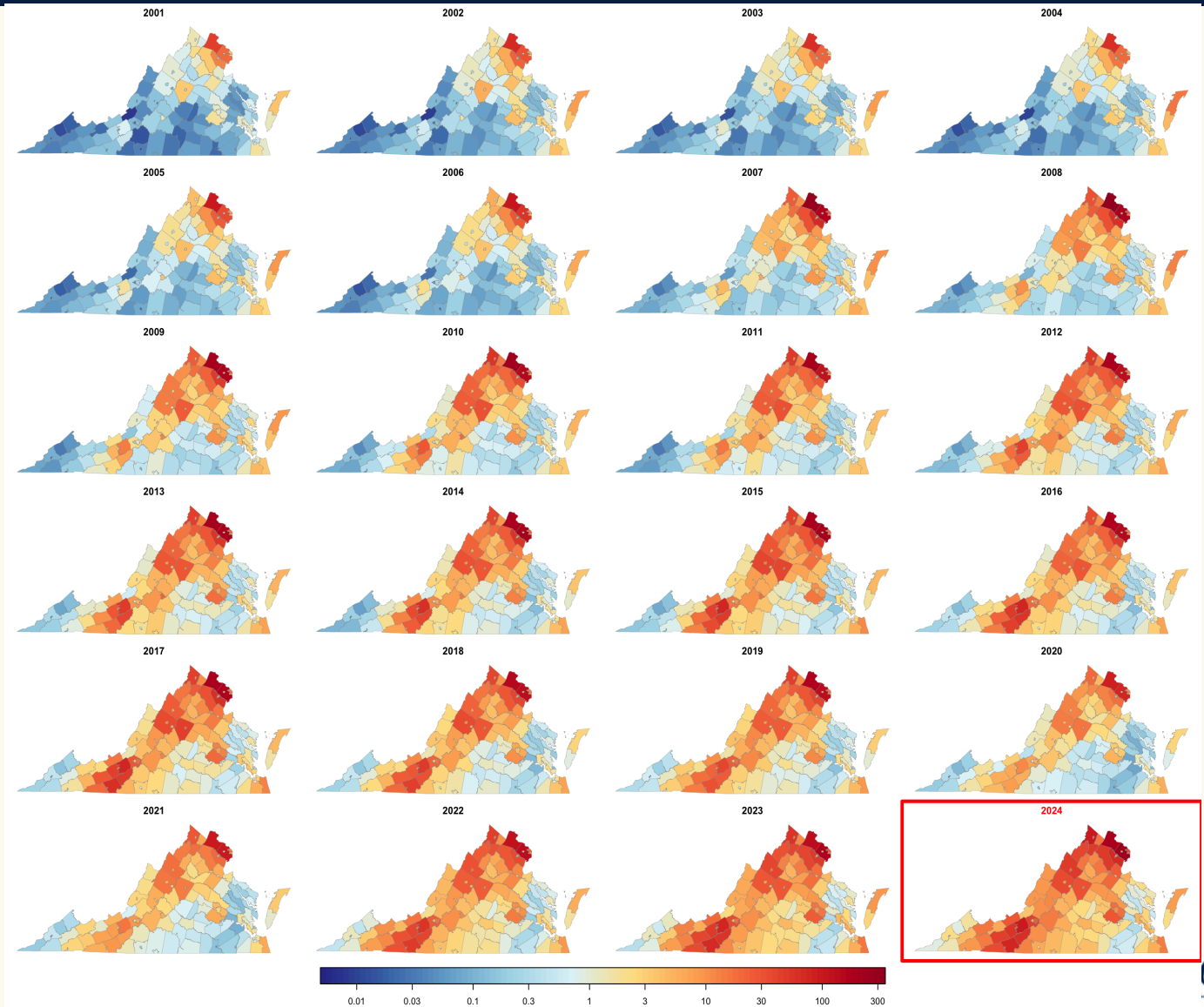
Table: Posterior results for the diffusion parameter of the PDE-based model

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Occurrence Probabilities Posterior Means



Posterior Means of μ



Results

Log-Linear Model:

- **Positive effects:** elevation, spring temperature, spring precipitation, winter precipitation, and spring wind v-component.
- **Negative effects:** spring humidity, fall humidity, spring wind u-component, fall wind v-component, summer wind u-component.

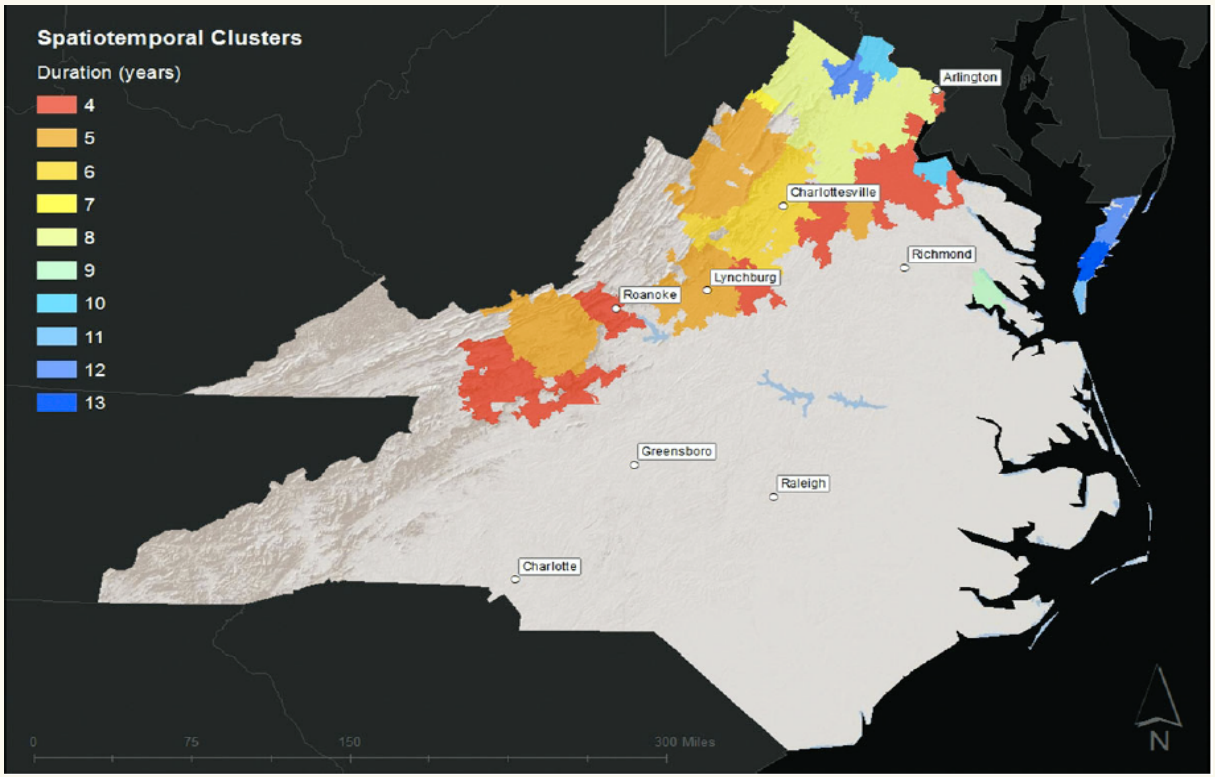
Logit model:

- **Positive effects:** None!
- **Negative effects:** spring wind v-component.

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Results

Our results on expansion of Lyme along the Appalachian Mountains corroborate other work:



From Lantos et al. (2015)

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Discussion

- Generally, larger number of cases in densely populated counties.
- Uptick in cases after 2007 (could be due to the expansion of the disease or also due to improved surveillance efforts)
- Since 2008, region of high activity in southwestern Virginia.
- Based on wind circulation results a possible hypothesis is that the disease vectors *may* be advected by the background flows toward the south in spring.

For a monthly level analysis of 2001-2016 data using a basic hurdle model (not PDE based), see:



Neupane, N., Goldbloom-Helzner, A., & Arab, A. (2021). Spatio-temporal modeling for confirmed cases of Lyme disease in Virginia. Ticks and Tick-borne Diseases.

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Forecasting

What about modeling & forecasting in the early stages of the spread of Lyme or in data sparse areas?

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Forecasting Disease Spread under Data Sparsity

Can we use information/knowledge from similar previous situations to forecast in the early stages of disease spread?

This may be done via **Transfer Learning**. One possible issue is the lack of timely data. Often, this may be addressed by using blended data (conventional and organic data).

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Transfer Learning

When performing transfer learning, we have target data denoted by $\{X_0, y_0\}$ and K source datasets denoted by $\{X_1, \ldots X_K\}$ and $\{y_1, \ldots, y_K\}$. Denote the number of observations in each dataset as N_k and the common number of variables as P.

Though transfer learning has been empirically extremely successful for different machine learning applications, statisticians have only recently begun to understand some of the unique theoretical properties of transfer learning in statistical contexts.

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Transfer Learning

Li et al. (2022) focused on transfer learning in the high dimensional linear case. In particular, they study the following two-step procedure:

$$\hat{\beta}_A \leftarrow \underset{\beta}{\operatorname{argmin}} \sum_{k=0}^{K} \|X_k\beta - y_k\|_2^2 + \lambda_A\|\beta\|_1 \tag{1}$$

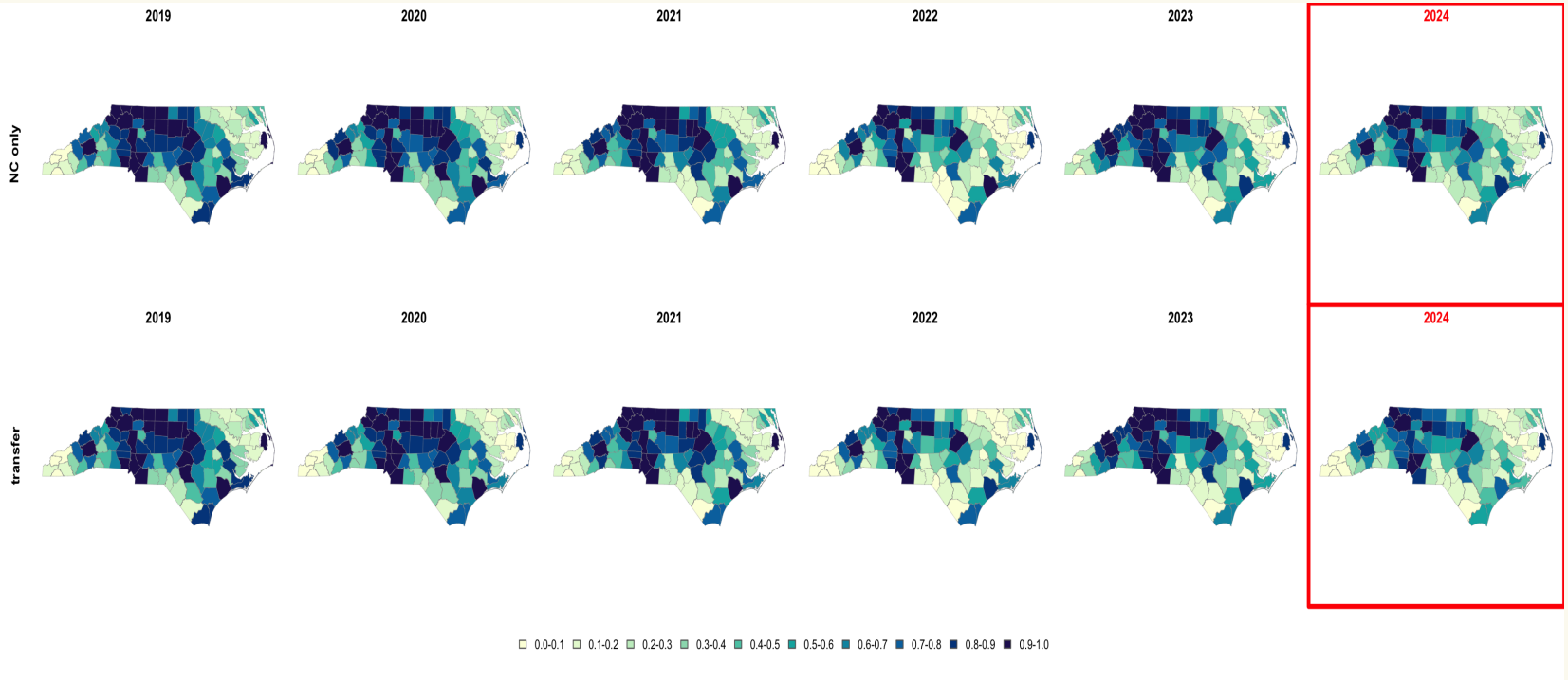
$$\hat{\delta}_T \leftarrow \underset{\delta}{\operatorname{argmin}} \|X_0\delta - (y_0 - X_0\beta_A)\|_2^2 + \lambda_\delta\|\delta\|_1 \tag{2}$$

Equation 1 performs a Lasso regression on all the data concatenated. Equation 2 performs a Lasso on the residuals of that regression on the target dataset. Then the regression coefficients for the target dataset are estimated as $\hat{\beta}_0 \leftarrow \hat{\beta}_A + \hat{\delta}_T$. Their method is known as **`Trans-Lasso`**.

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Transfer Learning for North Carolina

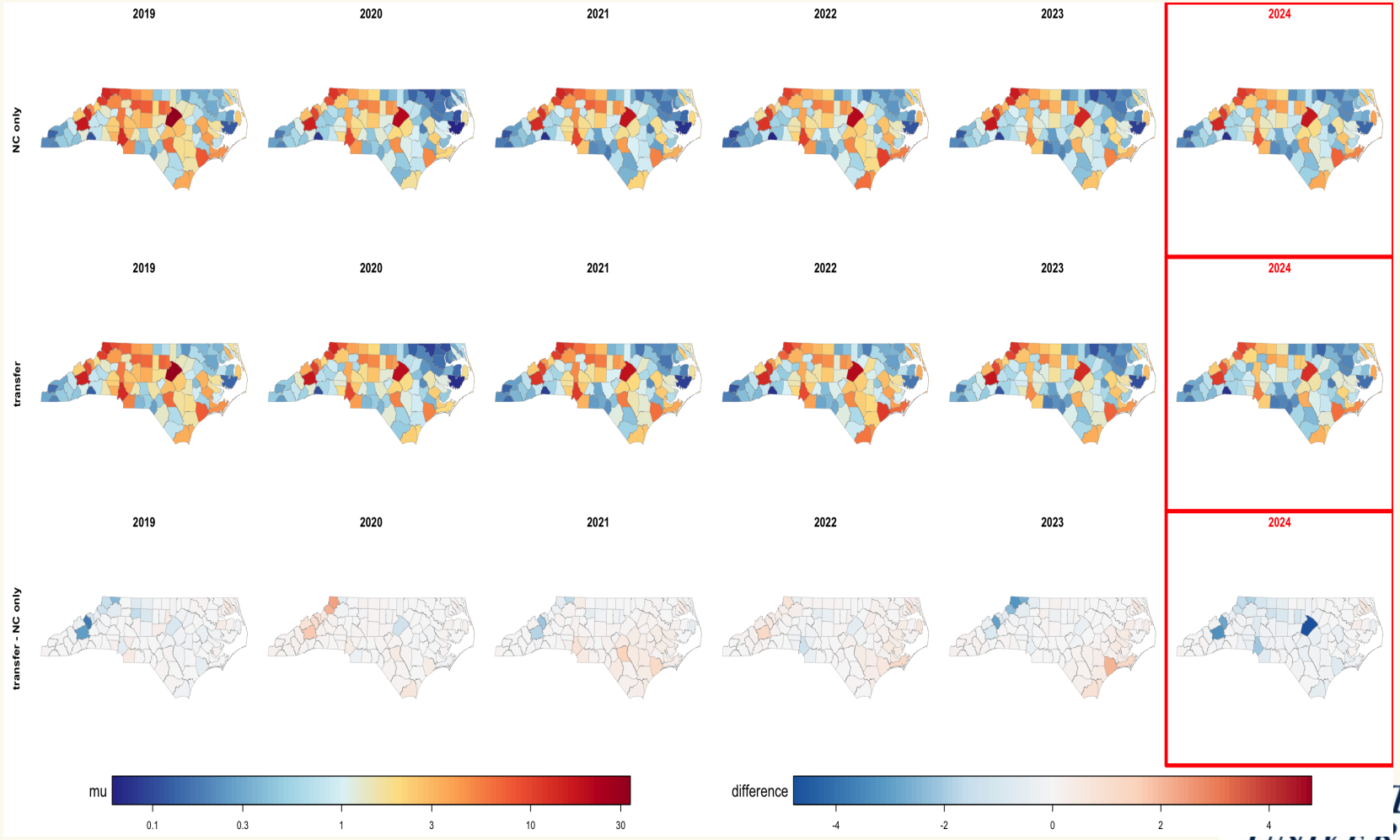
As an example, we applied transfer learning (Trans-Lasso on regression coefficients and transfer prior for θ) based on Lyme model results from Virginia to North Carolina.



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Transfer Learning for North Carolina

Posterior Means for μ



Transfer Learning for North Carolina

	RMSE		MAPE (%)		95% coverage (%)	
Year	NC only	Transfer	NC only	Transfer	NC only	Transfer
2019	**5.06**	5.42	74.2	**61.7**	97	92
2020	7.17	**6.97**	108.7	**90.8**	95	94
2021	**7.16**	7.24	**56.5**	58.8	91	90
2022	7.11	**7.03**	81.2	**65.6**	94	94
2023	**4.36**	4.42	72.3	**71.0**	95	94

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Bayesian Transfer Learning

An obvious way to perform transfer learning in the context of Bayesian statistics is to use the posterior conditioned on the source data as a prior for the target data. However, this leads to an overly concentrated posterior distribution.

It also does not address the issue of potential negative transfer.

Nathan Wycoff (UMass Amherst) and I (along with other collaborators) are currently working on a fully Bayesian TL approach.

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Future Work

- Better understand and use the drivers of dynamics. Patient-level data may provide additional insights.
- Use blended data (conventional and organic data) to inform the parameters governing the dynamics. In particular, tick surveys (Mostly based on citizen science projects) can provide important predictive information.
- Address bias issues related to organic data.
- Develop general Bayesian transfer learning methodology for complex count data models

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Acknowledgments

Earlier version of this work based on an autoregressive model for Lyme disease in Virginia is a joint work with Naresh Neupane and Ari Goldbloom-Helzner. [Ticks and Tick-borne Diseases, 2021]

The Bayesian transfer Learning work is based on collaboration with Lisa Singh (GU Computer Science/Massive Data Institute) and Katharine Donato (GU ISIM), and Nathan Wycoff (UMass Amherst). This project is supported by **NSF ATD-2428033**.

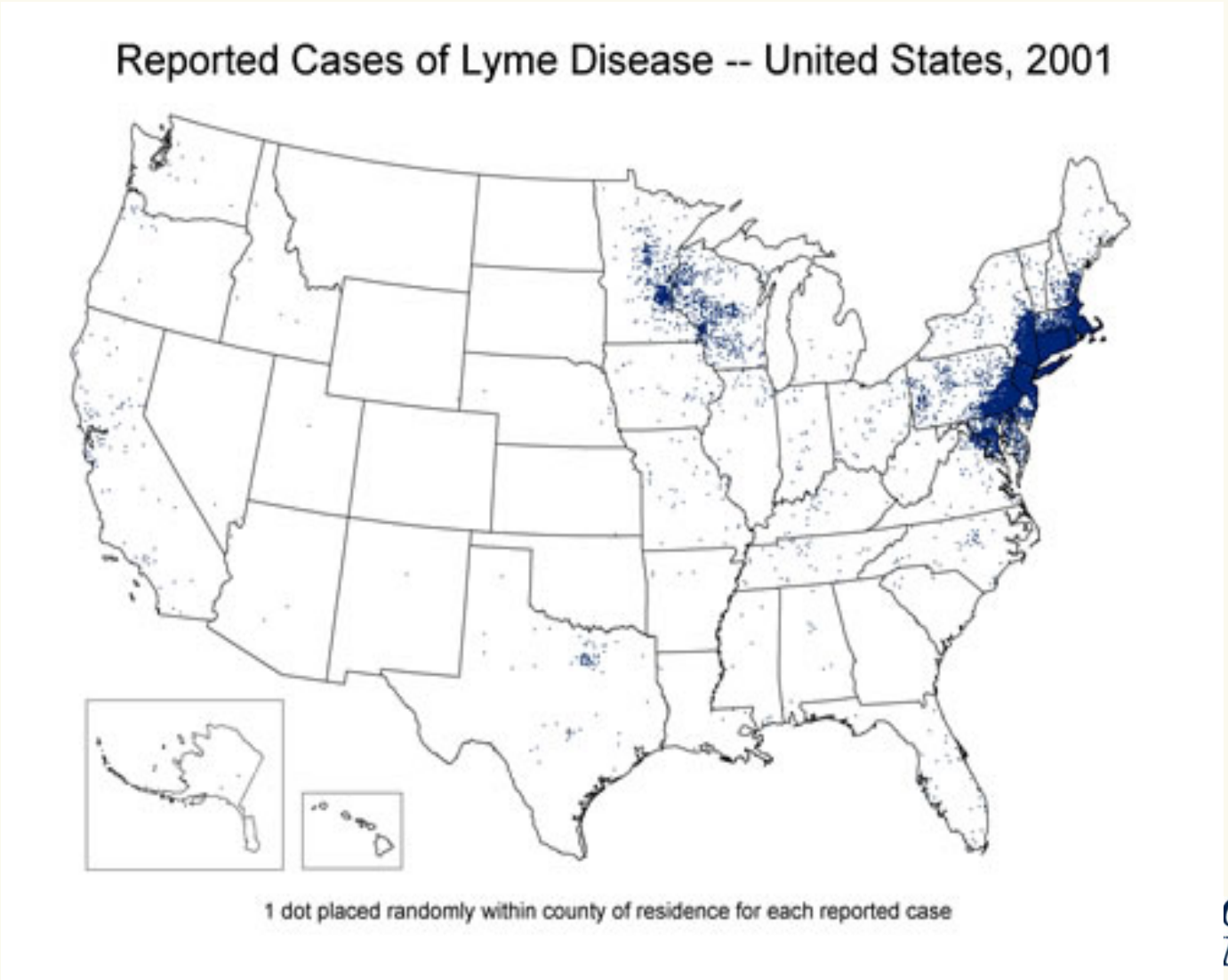
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References

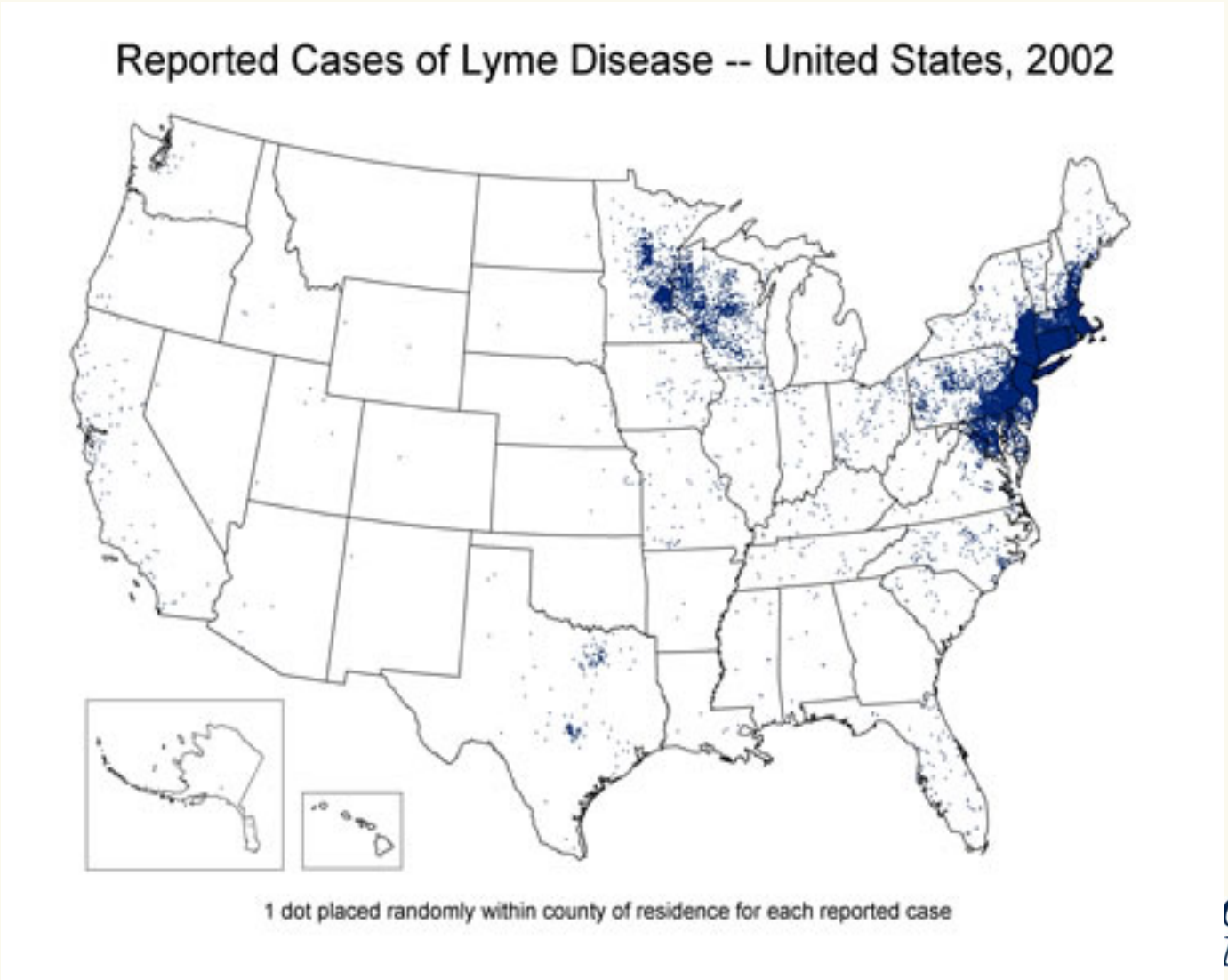
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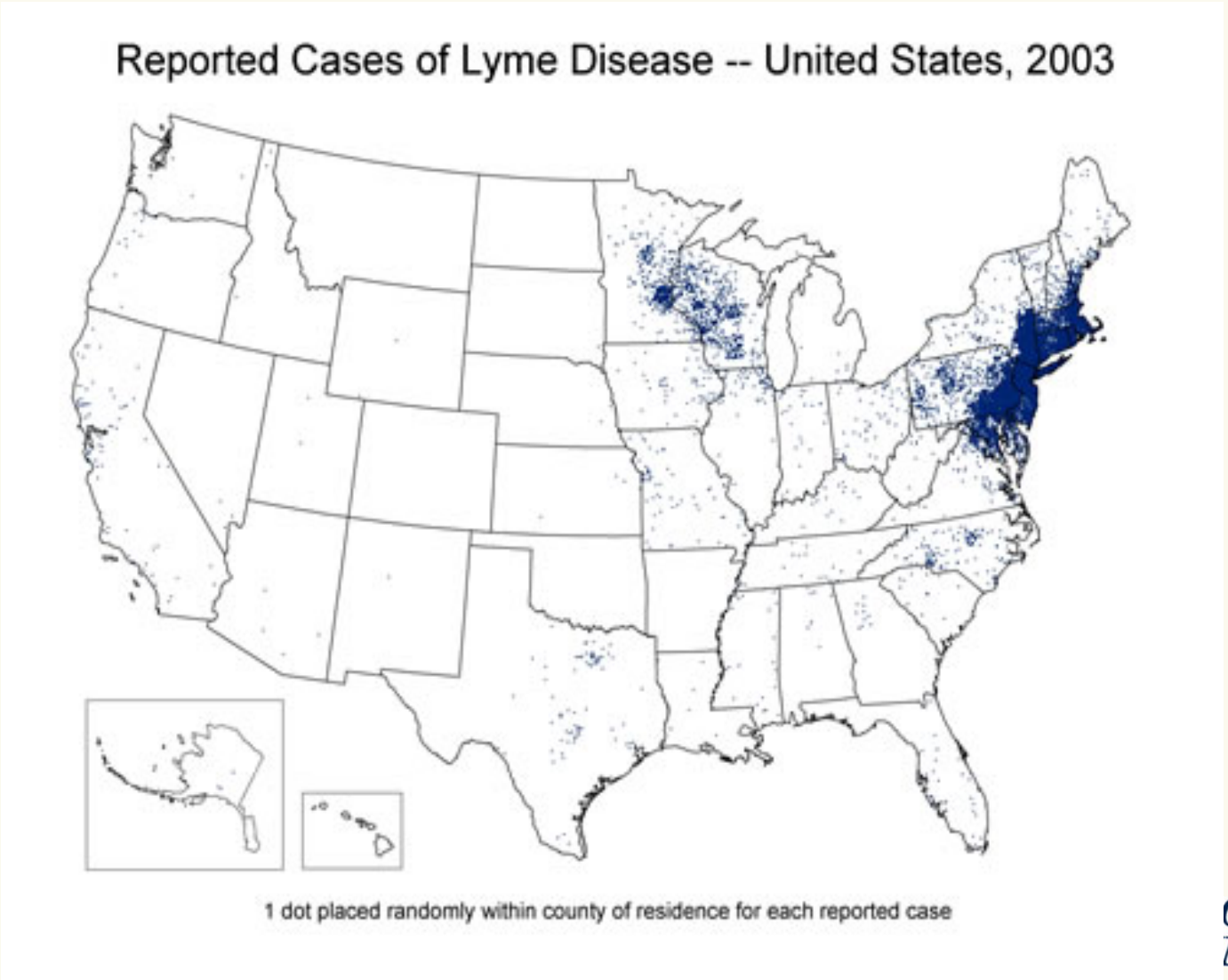
Lyme Disease



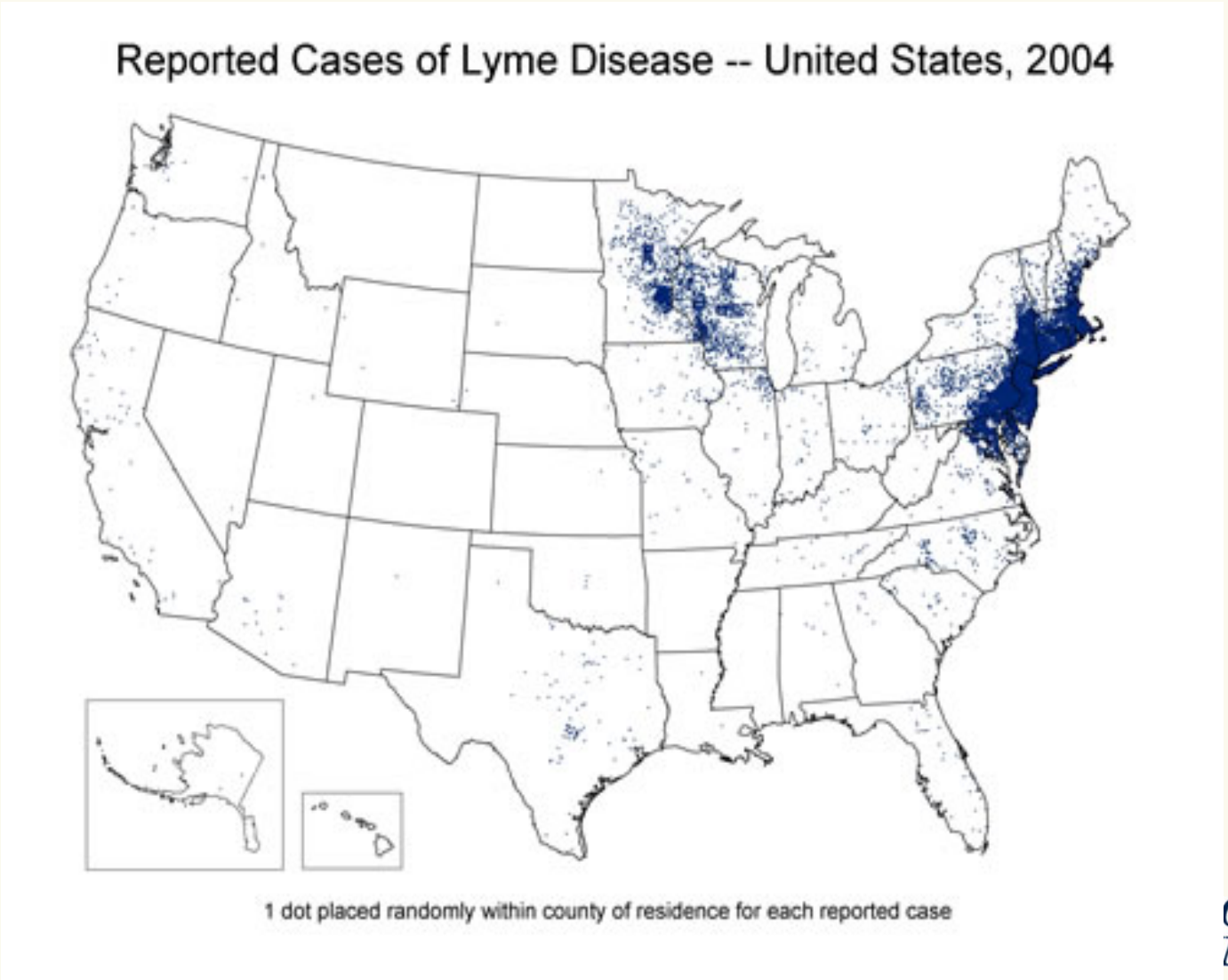
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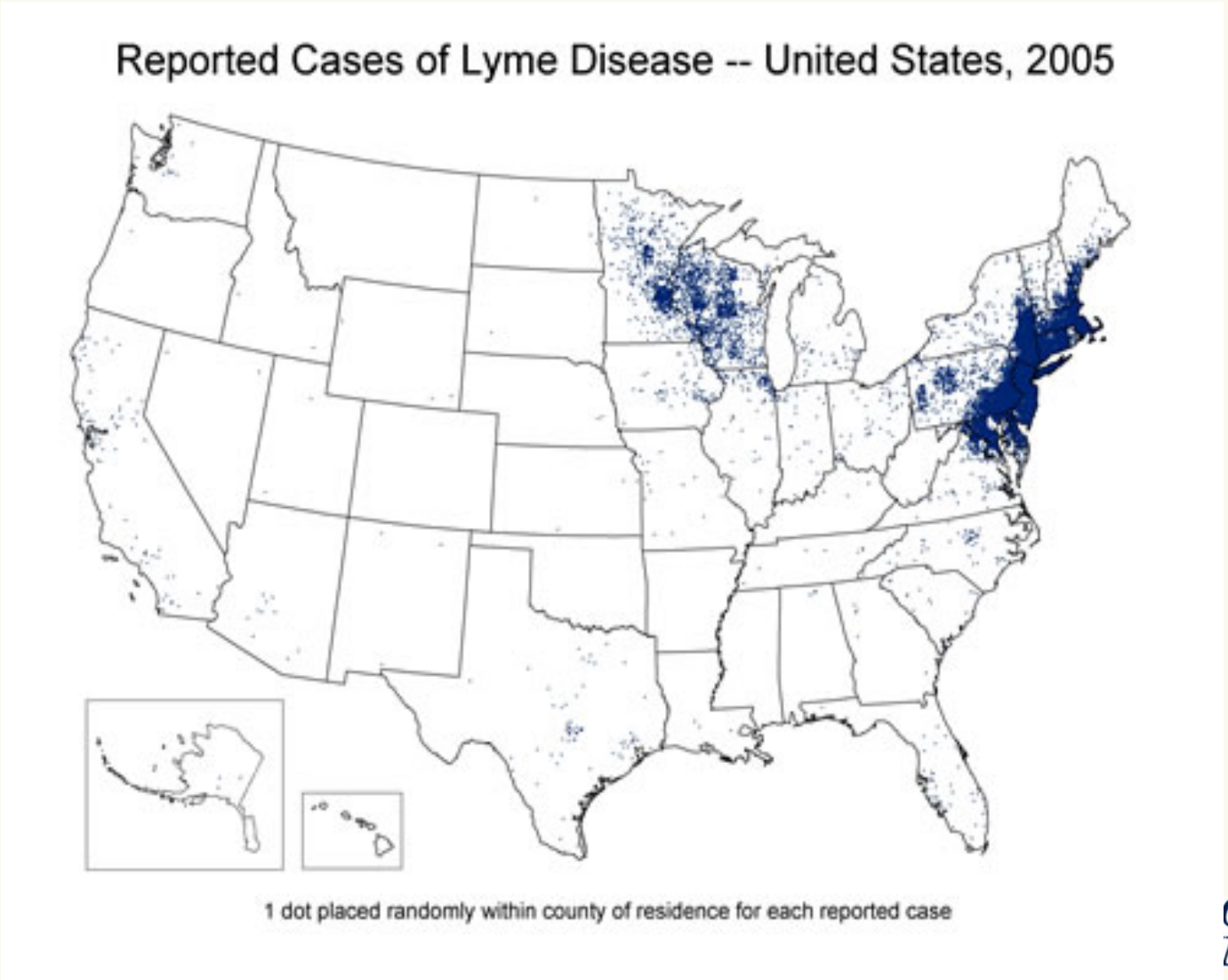
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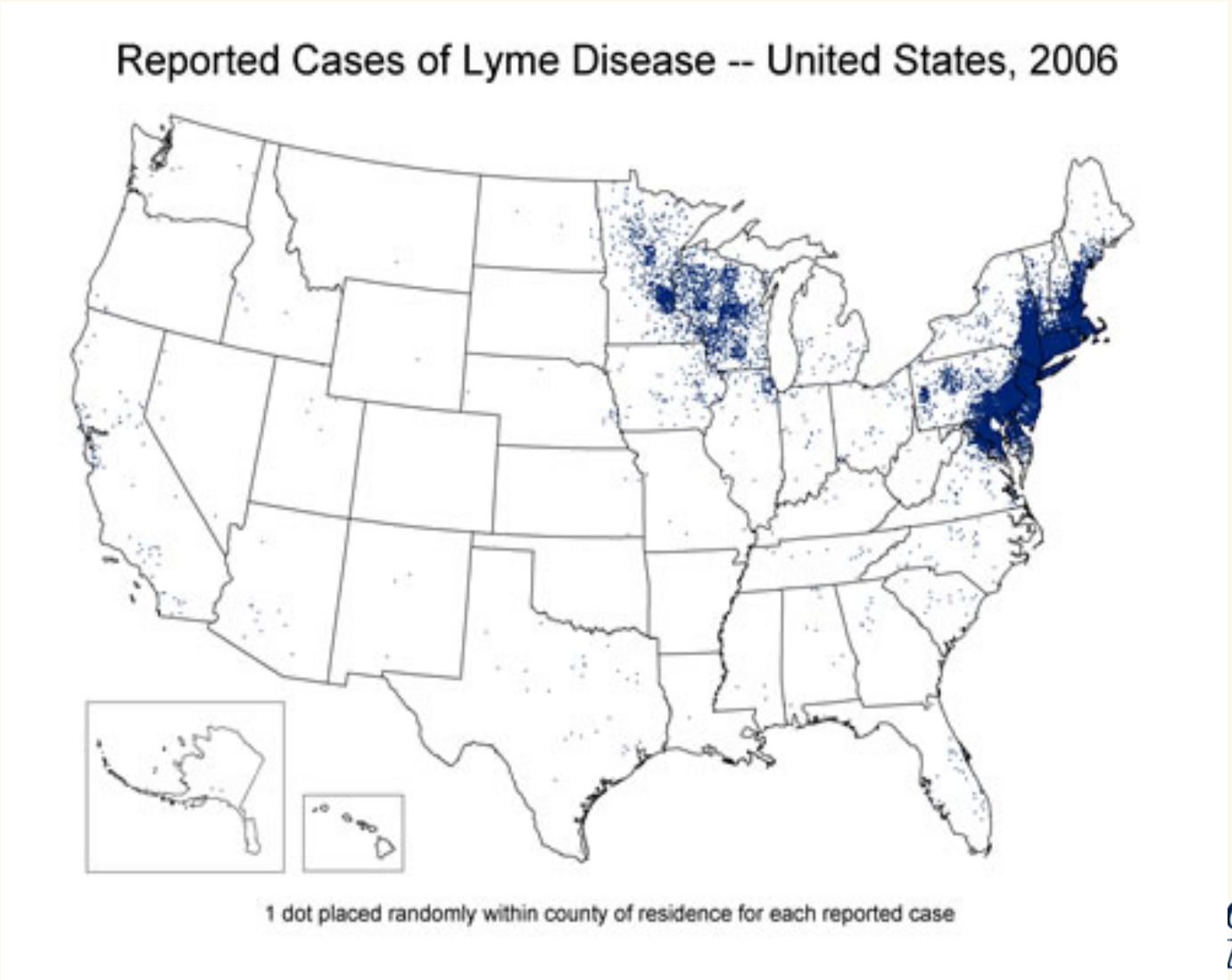
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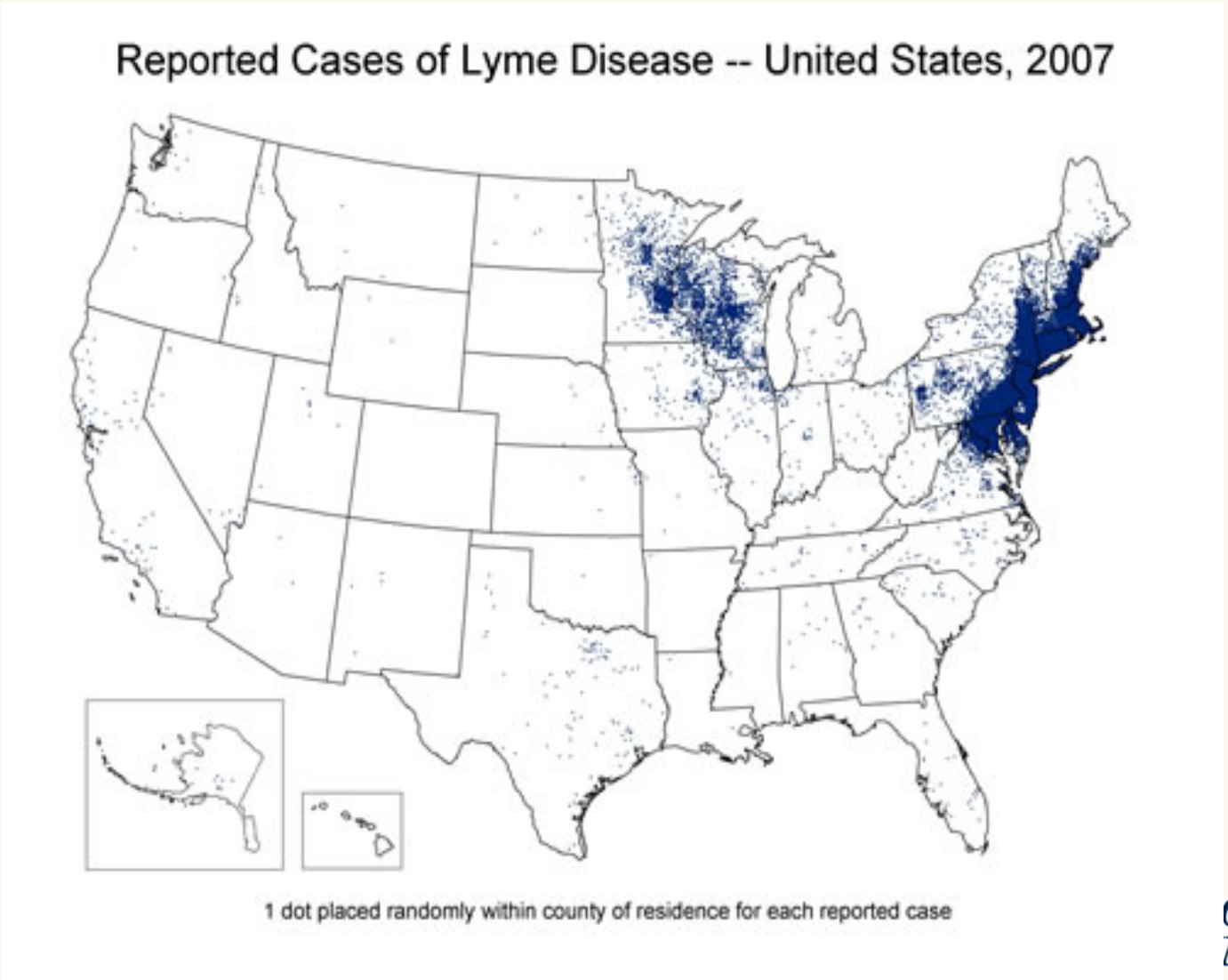
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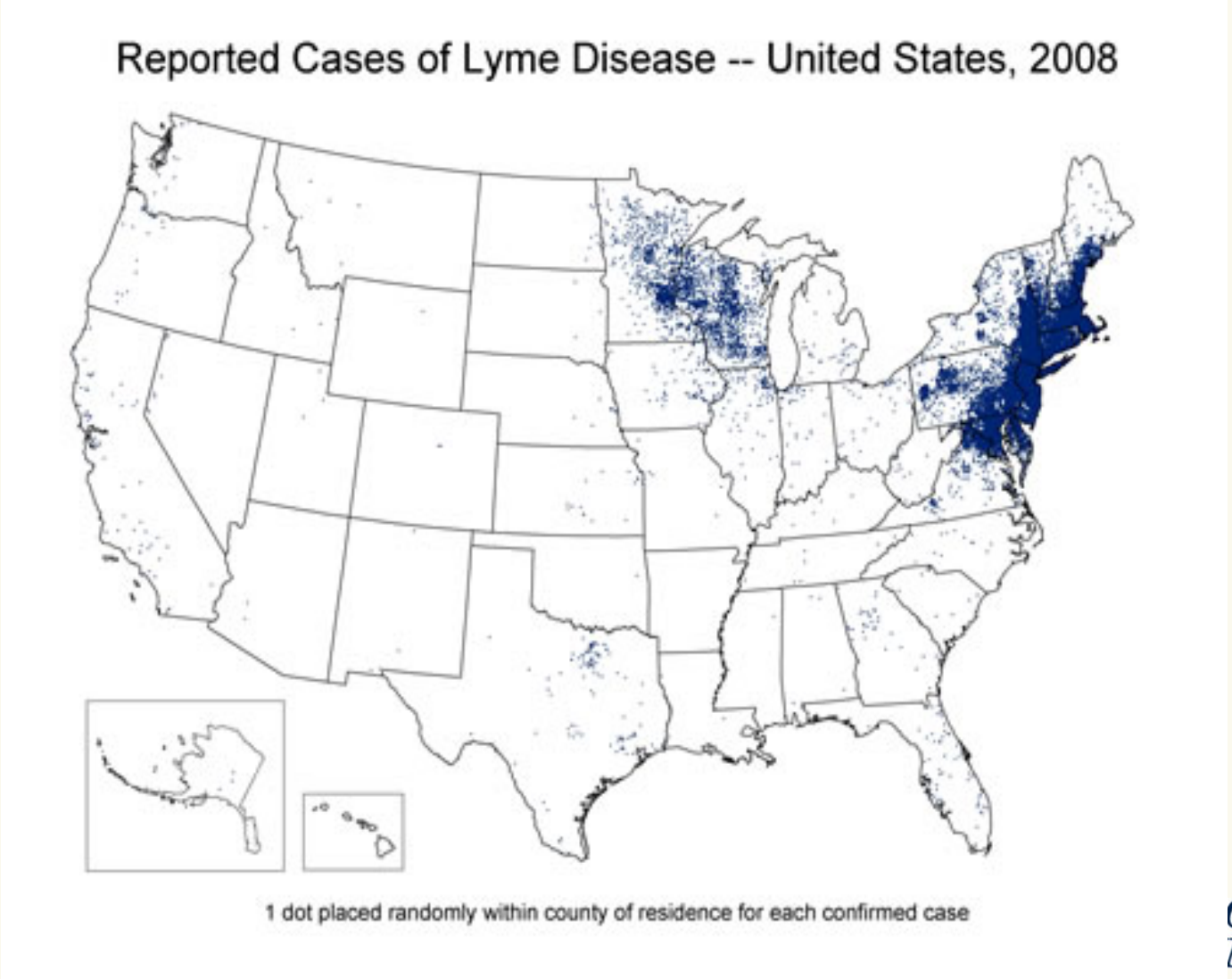
Lyme Disease



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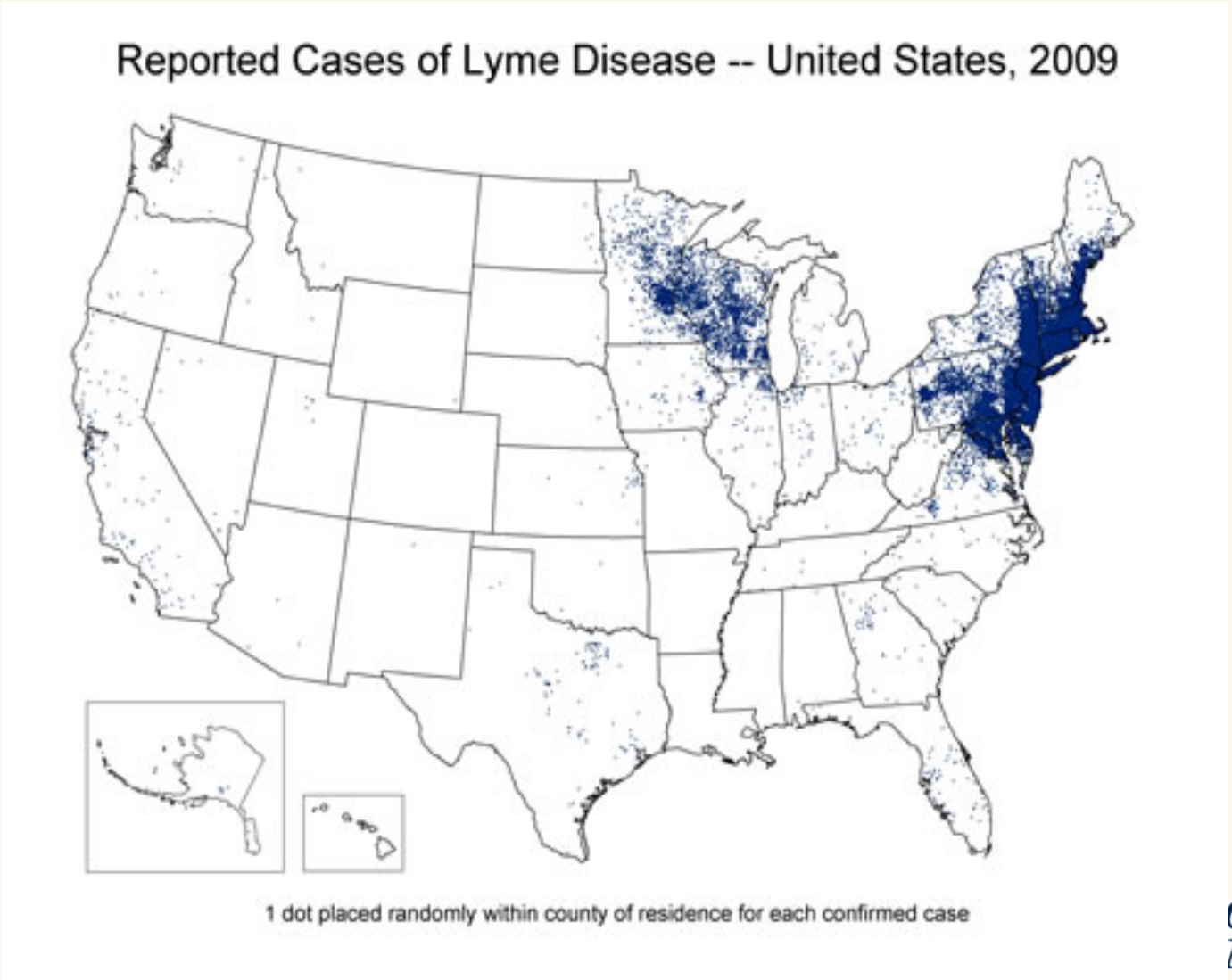


Lyme Disease

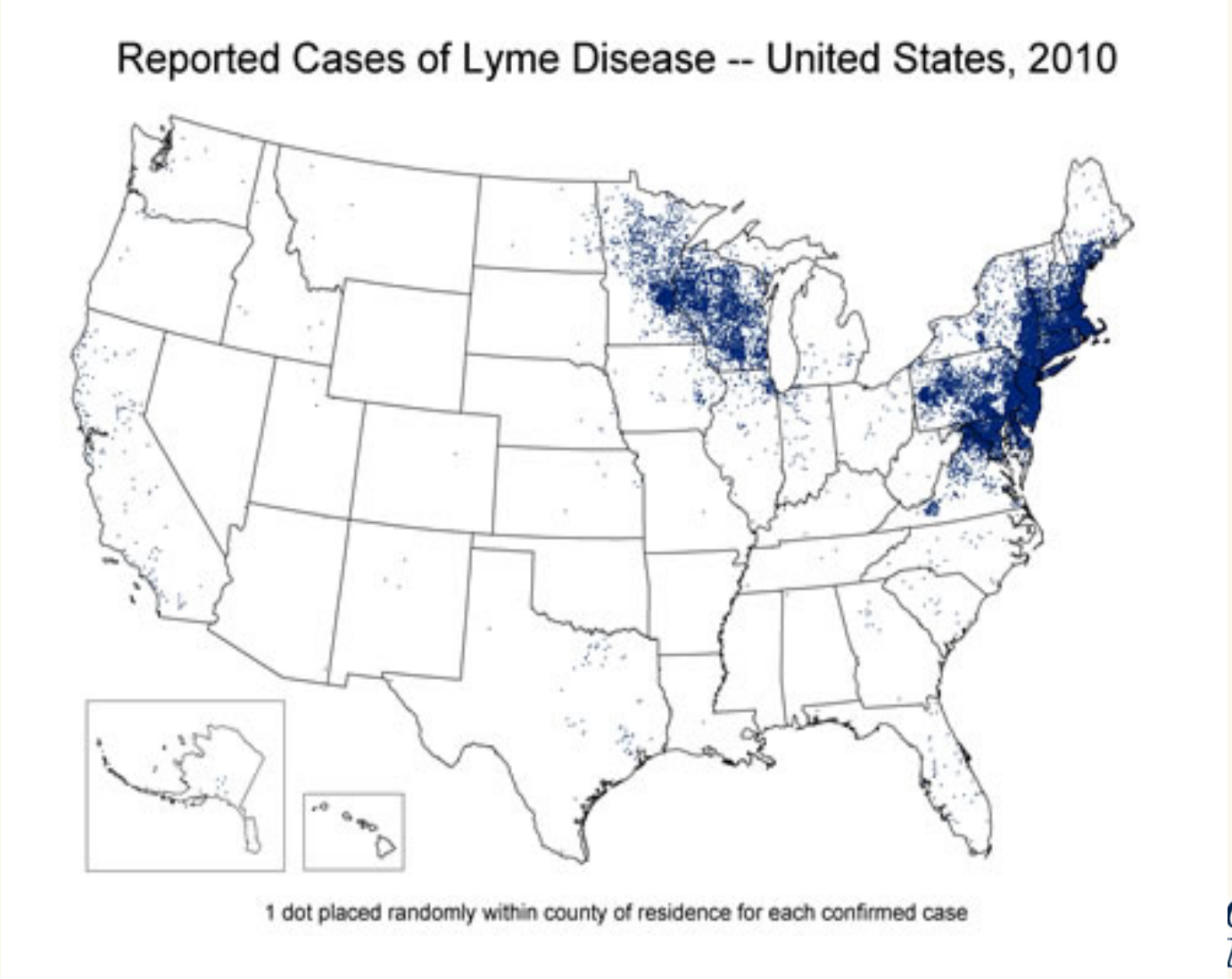


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Lyme Disease

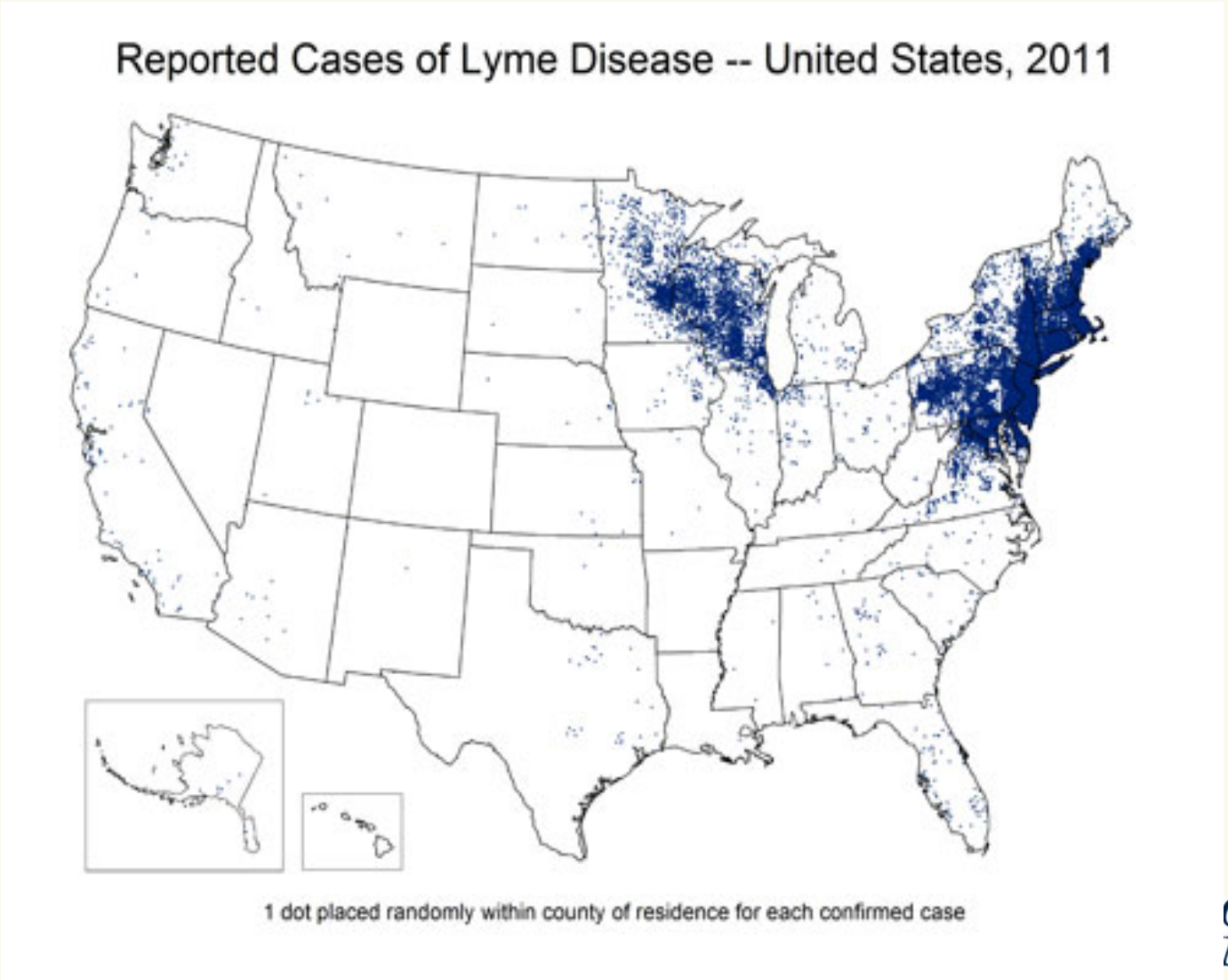


Lyme Disease



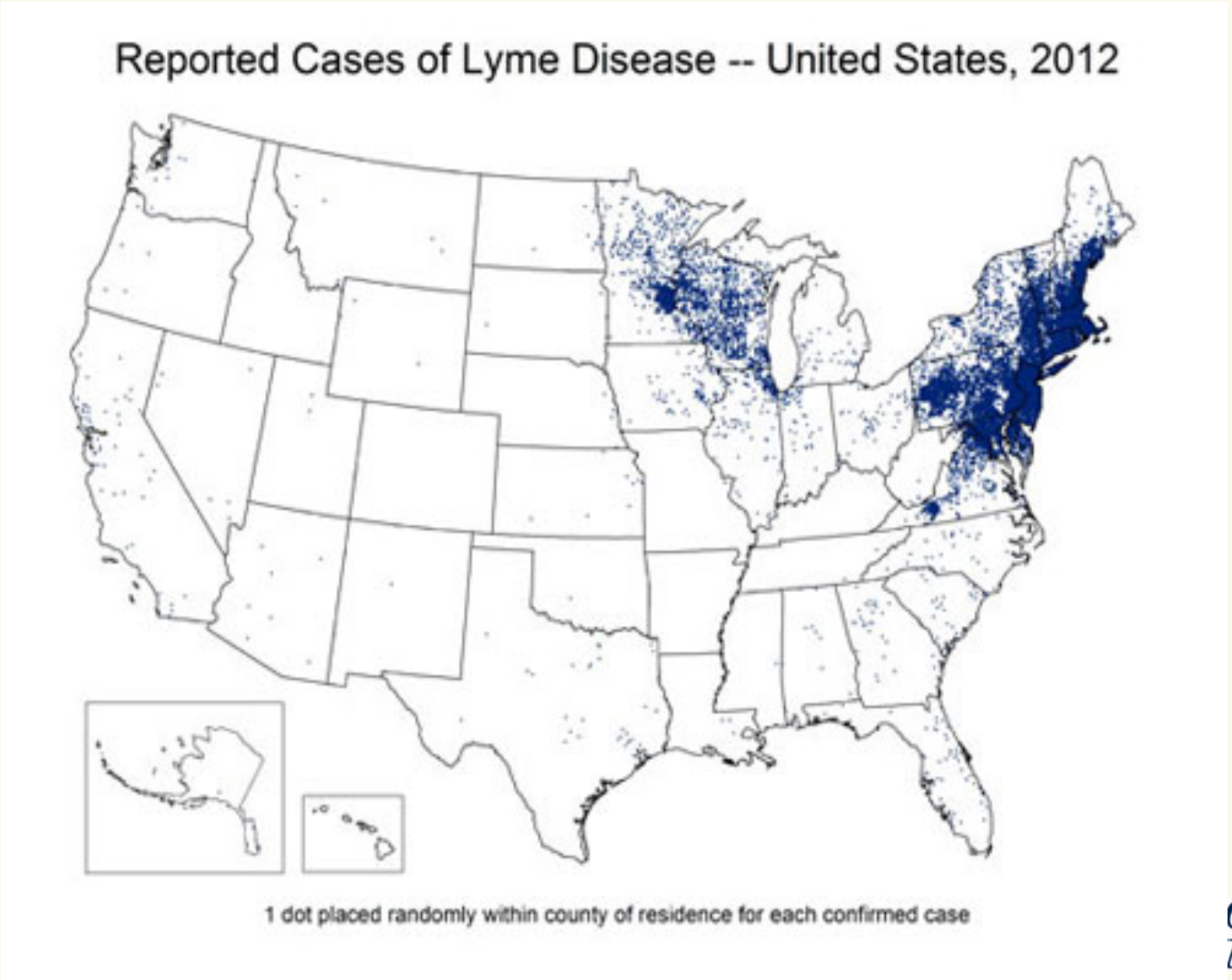
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Lyme Disease

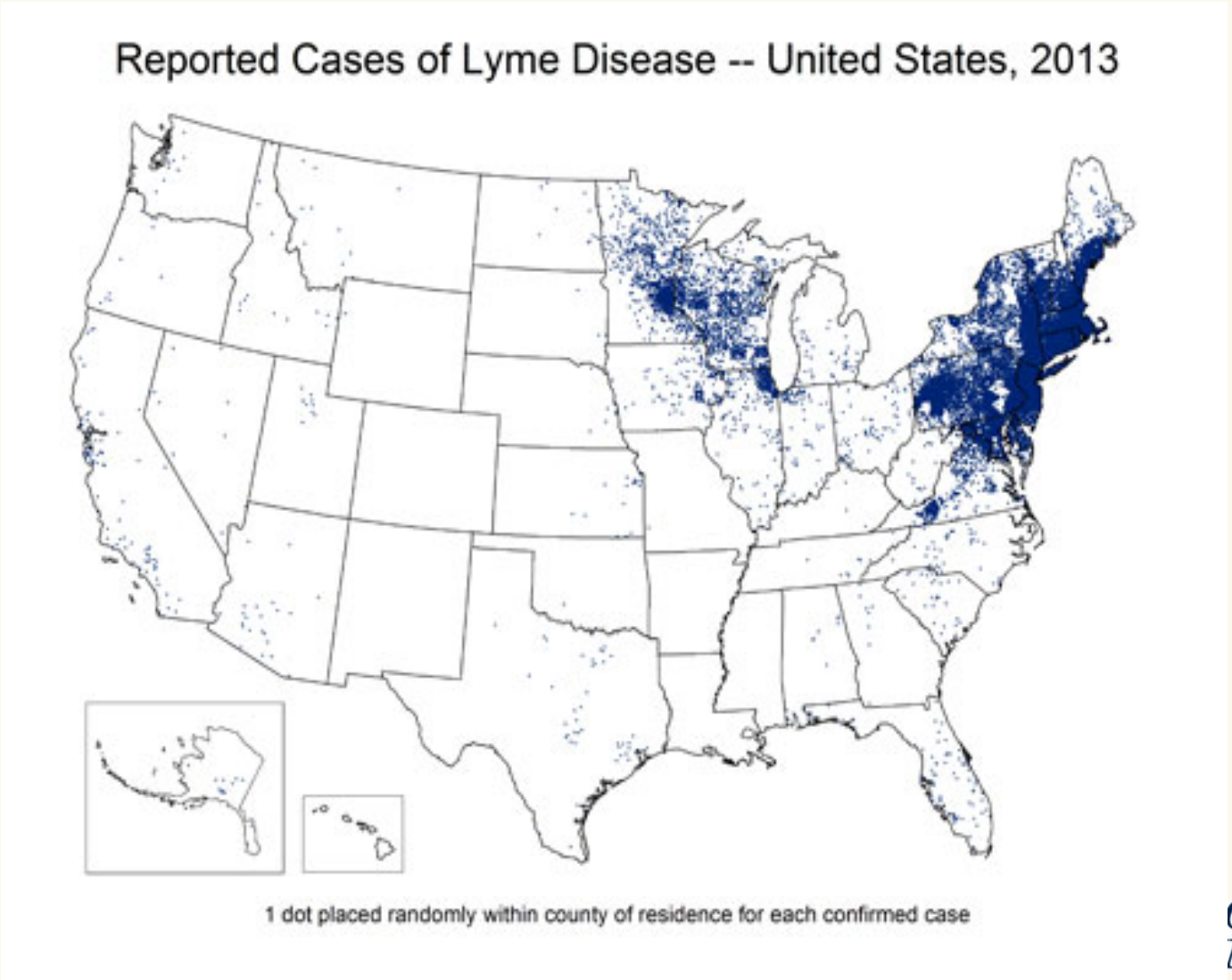


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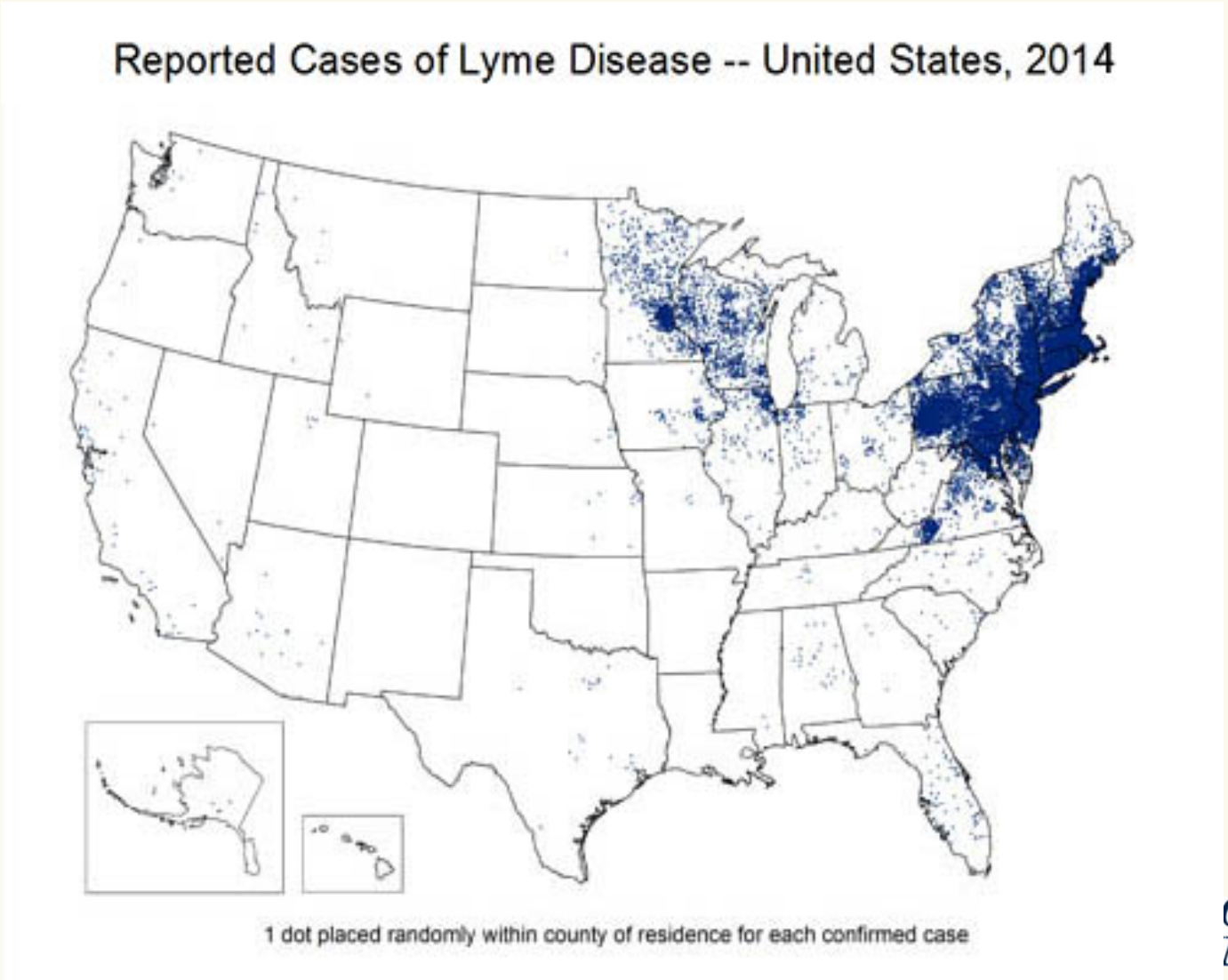
Lyme Disease



Lyme Disease

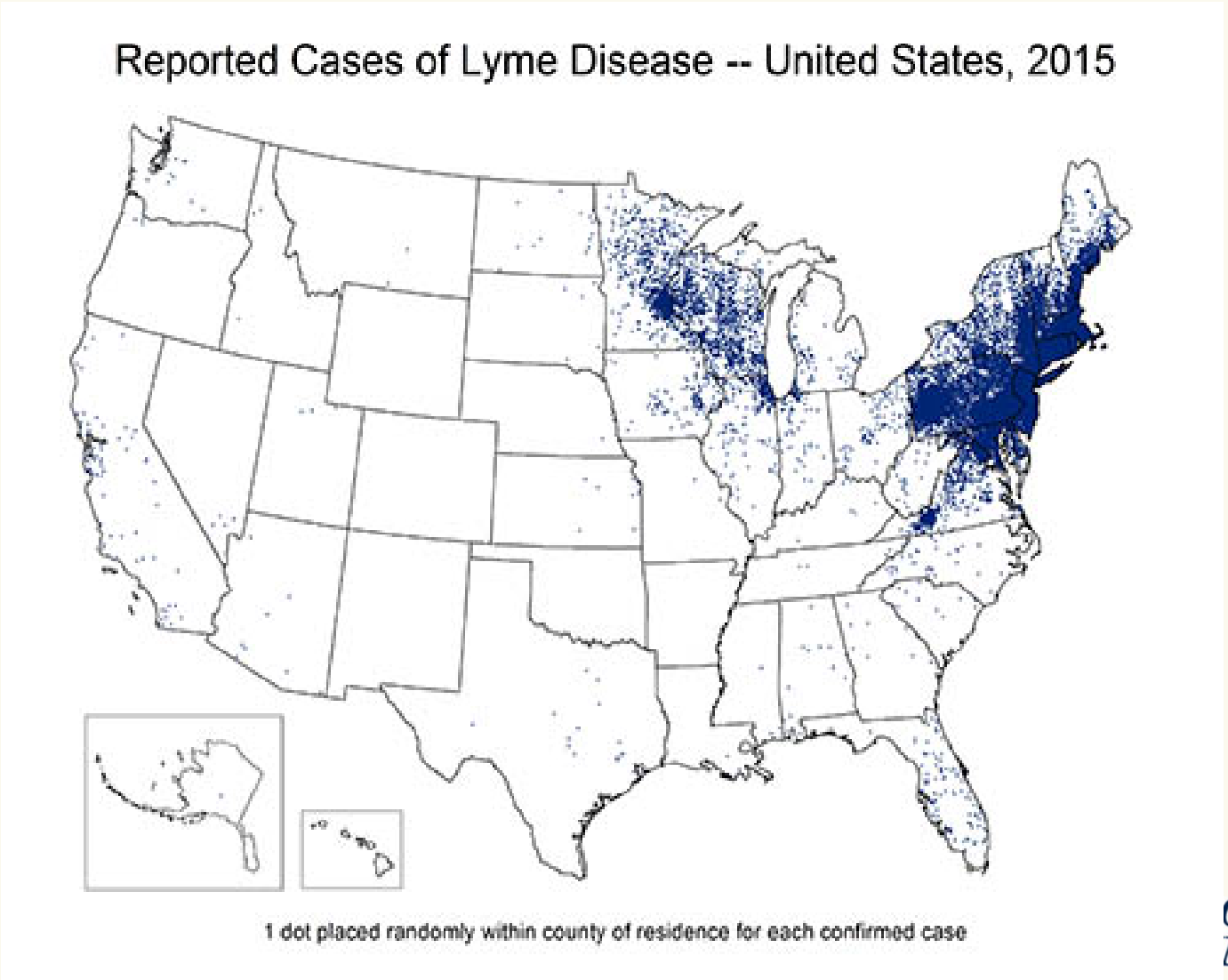


Lyme Disease

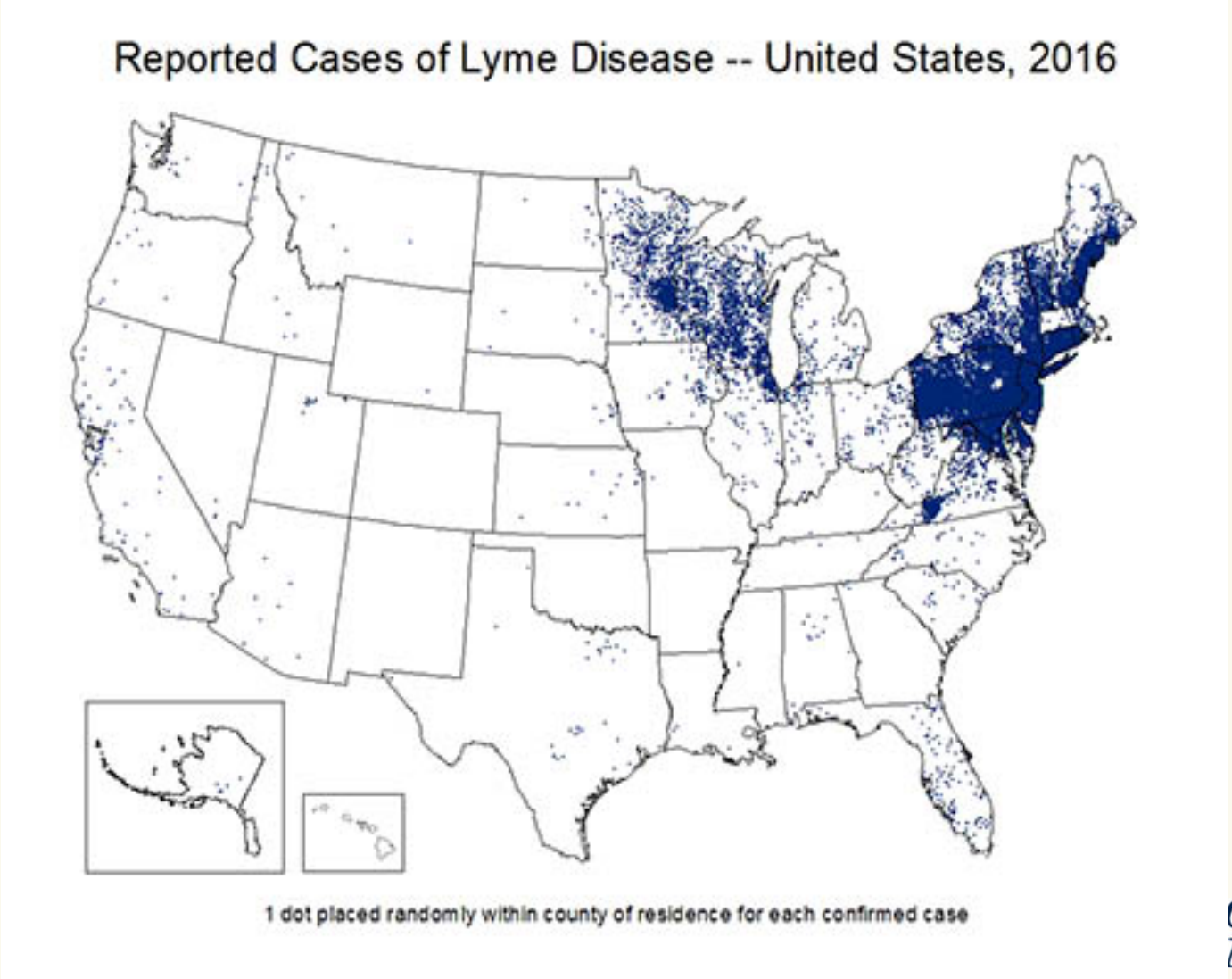


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Lyme Disease



Lyme Disease



Weather & Climate Effects?

Log-linear model Results:

Coefficient	Mean	SD	95% CI
January NDVI	-0.16	0.06	(-0.28, -0.03)
March NDVI	0.23	0.09	(0.06,0.39)
September NDVI	-0.27	0.10	(-0.47,-0.07)
April V-wind	0.28	0.06	(0.16,0.39)
May V-wind	-0.20	0.05	(-0.30,-0.11)
September V-wind	-0.19	0.05	(-0.29,-0.09)
August U-wind	-0.36	0.09	(-0.53,-0.19)
October U-wind	0.23	0.06	(0.11,0.35)
March Temperature	-0.39	0.11	(-0.60,-0.18)
January Relative Humidity	-0.17	0.07	(-0.30,-0.04)
March Relative Humidity	-0.33	0.11	(-0.53,-0.12)
July Relative Humidity	-0.29	0.10	(-0.49,-0.09)
September Relative Humidity	0.35	0.08	(0.19,0.51)
January Precipitation	-0.16	0.04	(-0.25,-0.08)
June Precipitation	-0.16	0.04	(-0.23,-0.08)
October Precipitation	-0.16	0.038	(-0.24,-0.09)
November Precipitation	0.09	0.04	(0.01,0.16)

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Weather & Climate Effects?

Logistic Regression Model Results:

Coefficient	Mean	SD	95% CI
February NDVI	0.48	0.16	(0.17, 0.79)
August V-wind	0.24	0.11	(0.03, 0.45)
October V-wind	0.60	0.16	(0.28, 0.92)
November V-wind	0.57	0.21	(0.16, 0.98)
December V-wind	-1.18	0.24	(-1.66, -0.72)
August U-wind	0.67	0.22	(0.24, 1.09)
October Precipitation	0.26	0.09	(0.09, 0.43)

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